A Weak-Form 2D Continuity Inversion for Pulsar Birth Demographics

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ABSTRACT

Inferring the initial spin periods of rotation-powered pulsars from one-dimensional (1D) period histograms or pulsar currents is ill-posed: dipole spin-down rates span six orders of magnitude across $(\log_{10} P, \log_{10} B)$, and flux-limited survey selection acts as an unknown visibility gauge. We present a non-parametric, weak-form two-dimensional (2D) continuity inversion in $(\log_{10} P, \log_{10} B)$ that transfers phase-space derivatives onto smooth test functions via integration by parts, eliminating numerical differentiation noise on sparse catalogues. To prevent circularity in $P_0 \in [0.40, 0.85]$ s, the visibility gauge is calibrated exclusively outside this target window. Across 100-seed synthetic benchmarks ($N = 2,356$), the inversion accurately recovers fast, Gaussian, and bimodal birth densities with per-seed Wasserstein errors $W_1(\log_{10} P_0) = 0.047$–$0.057$ dex and slow-window resolution FWHM $= 0.29$ dex. Calibrating modified frequentist CL_s exclusion curves across 300-seed Monte Carlo ensembles yields conservative 95% upper limits on the slow-birth fraction $f_{\rm slow}^{\rm birth}$ of 25%–30% (P1–P4; weakening by 5–10 pp under profiled gauges), alongside an apparent fraction at the resolution limit of 33%–41% under an independent 3D forward model (13.7% floor). Applied to $N = 2,356$ isolated ATNF pulsars, the inversion yields $\hat{f}_{\rm slow}^{\rm birth} = 0.57\%$ ($\leq 9.03\%$ across all field decay rates), corroborated by four gauge-free cross-checks: a supernova remnant census ($f_{\rm slow}^{\rm birth} \leq 22\%$), a Galactic scale-height kinematic clock (2.6σ, $\leq 17\%$), volume-complete cylinders ($\leq 4\%$–9%), and a high-B strip conservation bound ($\leq 23\%$). The classical $\geq 40\%$ injection hypothesis is excluded across all protocols; slow-birth fractions below $\sim 25\%$–30% lie below the current resolution limit.

1. INTRODUCTION

1.1. *Background: The 1D Pulsar Current and the "Injection" Debate*

Core-collapse conservation of angular momentum predicts that newborn neutron stars spin rapidly at birth ($P_0 \sim$ 10–100 ms; Ostriker & Gunn 1969; Goldreich & Julian 1969), consistent with young historical supernova remnant (SNR) pulsars such as the Crab ($P =$ 33.4 ms) and Vela ($P = 89.3$ ms). However, Vivekanand & Narayan (1981) introduced a demographic diagnostic known as the one-dimensional (1D) *pulsar current*, $J(P) \equiv \dot{P}\, n(P)$, which measures the net flow of pulsars crossing spin period P. Under steady-state conservation ($\partial_t n = 0$), the 1D continuity equation along the period axis is $\partial_P J(P) = S_{\rm 1D}(P) - D_{\rm 1D}(P)$, where $S_{\rm 1D}(P) \geq 0$ and $D_{\rm 1D}(P) \geq 0$ denote the 1D birth and death rates per unit period. If all pulsars are born at short periods ($P_0 < 0.1$ s) and no pulsars turn off before $P \sim 1$ s, $J(P)$ must remain flat on $P \in [0.1, 1.0]$ s. Instead, Vivekanand & Narayan (1981) found that the empirically weighted current rose steeply between $P \approx 0.1$ s and $P \approx 0.5$ s, concluding that 40%–75% of all Galactic pulsars are "injected" into the active population at slow periods, $P_0 \in [0.40, 0.85]$ s, either because they are born spinning slowly or because their surface dipole magnetic fields emerge only after a delay of $\tau_{\rm emerge} \sim 10^5$–$10^6$ yr. We note at the outset that this delayed-field-emergence scenario is not tested by our framework, which evolves each pulsar from a fixed initial dipole field B_0 at $t = 0$.

Subsequent parametric and 1D current studies supported a large slow-born sub-population (Chevalier & Emmering 1986; Narayan 1987; Emmering & Chevalier 1989; Narayan & Ostriker 1990), whereas Lorimer et al. (1993) showed that Poisson fluctuations and flux-luminosity selection biases can generate artificial rises in binned 1D currents. In a foundational three-dimensional (3D) forward Monte Carlo synthesis of the Parkes Multibeam Pulsar Survey (Manchester et al. 2001), Faucher-Giguère & Kaspi (2006) (hereafter FGK06) reproduced the observed $(P, \dot{P}, D, S_{1400})$ distribution without a bimodal injection peak using a single unimodal Gaussian initial period distribution with mean $\langle P_0 \rangle = 0.30$ s and standard deviation $\sigma_{P_0} = 0.15$ s truncated at $P_0 > 0$:

$$p_{\rm FGK06}(P_0) = \frac{\Theta(P_0)}{\Phi(\langle P_0 \rangle / \sigma_{P_0}) \sqrt{2\pi}\, \sigma_{P_0}} \times \exp\left[-\frac{(P_0 - \langle P_0 \rangle)^2}{2\sigma_{P_0}^2} \right], \tag{1}$$

where $\Phi(z)$ is the standard normal cumulative distribution function ($\Phi(2) = 0.97725$). Integrating Equation (1) across $P_0 \in [0.40, 0.85]$ s yields a global slow-birth tail of

$$f_{\rm slow}^{\rm FGK06} = \frac{\Phi\left(\frac{0.85-0.30}{0.15}\right) - \Phi\left(\frac{0.40-0.30}{0.15}\right)}{\Phi(0.30/0.15)} = \frac{0.99988 - 0.74751}{0.97725} = 25.82\%. \tag{2}$$

Coupled magneto-thermal syntheses incorporating Hall drift and Ohmic dissipation (Aguilera et al. 2008; Popov et al. 2010; Viganò et al. 2013; Gullón et al. 2014) further demonstrated that P_0 is broadly degenerate with magnetic field evolution for $P_0 \lesssim 0.3$–0.5 s, while inverse reconstructions on characteristic-age subsamples

(Igoshev & Popov 2013) highlighted systematic biases when field decay and survey selection are coupled. Note that $f_{\rm slow}^{\rm FGK06} = 25.8\%$ in Equation (2) is a parametric point prediction for the global birth-fraction tail ($f_{\rm slow}^{\rm birth}$ defined below); window-averaged ($f_{\rm slow}^{\rm window}$) and exit-boundary ($f_{\rm slow}^{\rm exit}$) diagnostics probe different physical quantities and survey sub-populations and are therefore not numerically comparable to $f_{\rm slow}^{\rm FGK06}$, while frequentist CL_s bounds on $f_{\rm slow}^{\rm birth}$ remain conditioned on the assumed forward operator.

The contributions of this paper are organized around the inverse methodology:

1. *Weak-form 2D continuity solver and resolution theory*: We formulate the 2D phase-space continuity inverse problem in $(\log_{10} P, \log_{10} B)$ in weak (variational) form, characterize its singular spectrum, effective degrees of freedom, and resolution matrix $\mathbf{R}$, and introduce a non-circular conditional likelihood that calibrates the visibility gauge strictly outside the $[0.40, 0.85]$ s test window.

2. *Monte Carlo dose-response and CL_s exclusion framework*: We calibrate the exact finite-sample dose-response slope, 0%-dose boundary floor, and modified frequentist CL_s exclusion power across 300-seed ensembles under matched gauges, functional visibility misspecification, death-line scatter, and blind inversions of an independent 3D FGK06 synthesis code.

3. *Application to the ATNF catalogue and gauge-free cross-checks*: We apply the calibrated 2D inversion to $N = 2,356$ isolated ATNF pulsars across a branch-free magnetic field decay (h_0) scan and cross-check the result against four observational diagnostics that require no fitted 2D visibility gauge.

1.2. *Definitions of Slow-Birth Estimands and Core Methodological Obstacles*

Because pulsar spin-down residence times depend strongly on both birth period P_0 and magnetic field B_0, different demographic diagnostics measure mathematically distinct slow-birth fractions. Throughout this paper we distinguish three specific estimands on the target interval $[P_1, P_2] = [0.40, 0.85]$ s ($x \equiv \log_{10}(P/\mathrm{s}) \in [-0.398, -0.070]$, $y \equiv \log_{10}(B/\mathrm{G})$):

$$
\begin{aligned}
f_{\rm slow}^{\rm birth} &\equiv \frac{\iint_{\mathcal{W}} S(x_0, y_0)\, dx_0\, dy_0}{\iint S(x_0, y_0)\, dx_0\, dy_0} \\
&\quad \text{(Birth fraction in } \mathcal{W} \equiv [0.40, 0.85] \text{ s)}, \\
f_{\rm slow}^{\rm window} &\equiv \frac{N_{\rm slow,vis}(\mathcal{W})}{N_{\rm tot,vis}(\mathcal{W})} \\
&\quad \text{(Detected fraction born in } \mathcal{W}), \\
f_{\rm slow}^{\rm exit} &\equiv \frac{J_{\rm slow}(P_2 = 0.85 \text{ s})}{J_{\rm tot}(P_2 = 0.85 \text{ s})} \\
&\quad \text{(High-}B \text{ strip exit current fraction)}.
\end{aligned}
\tag{3}
$$

Here $f_{\rm slow}^{\rm birth}$ is the unweighted fraction of all newborn pulsars injected into $P_0 \in [0.40, 0.85]$ s (the quantity constrained by our 2D PDE inversion, the CL_s exclusion curves, the SNR census, and Equation 2); $f_{\rm slow}^{\rm window}$ is the residence-time-weighted fraction of currently detected pulsars inside $P \in [0.40, 0.85]$ s that were born inside that window (probed by the kinematic scale-height clock); and $f_{\rm slow}^{\rm exit}$ is the fraction of the pulsar current exiting $P_2 = 0.85$ s within an active high-B strip that originated from births inside $[0.40, 0.85]$ s (probed by the strip conservation inequality).

Constraining these quantities requires overcoming three methodological obstacles:

1. *Two-dimensional kinematic mixing*: Because dipole spin-down obeys $\dot{P} \propto B^2/P$ across four decades of magnetic field ($10^{10.5}$–$10^{13.5}$ G), pulsars at the same period $P = 0.5$ s have spin-down rates $\dot{P}$ spanning six orders of magnitude. Projecting the 2D (P, B) plane onto 1D period bins mixes young, fast-decelerating high-B pulsars with old, slowly decelerating low-B pulsars piled up near the pair-cascade death line.

2. *Observational truncation and the visibility gauge degeneracy*: In flux-limited surveys, the observed phase-space density $f_{\rm obs}(P, B) = \theta_{\rm vis}(P, B)\, f_{\rm true}(P, B)$ is multiplied by an effective detection probability $\theta_{\rm vis}(P, B)$ combining Malmquist flux selection (Schmidt 1968; Lynden-Bell 1971) and geometric beaming $f_{\rm beam}(P)$ (Lyne & Manchester 1988; Rankin 1993; Tauris & Manchester 1998). If $\theta_{\rm vis}(P, B)$ is fitted across $P \in [0.40, 0.85]$ s under a zero-birth assumption, the gauge can absorb a genuine slow-birth signal; conversely, unmodeled 3D selection effects can leak into the inferred source term.

3. *Boundary clamping and regularization attenuation*: Enforcing physical non-negativity ($S(x_0, y_0) \geq 0$) on a regularized inverse problem with finite sample size ($N \sim 2,000$) produces a right-skewed 0%-dose distribution and smooths localized injections across grid boundaries. Point

estimates alone are insufficient without a calibrated dose-response and frequentist exclusion curve.

1.3. *Outline of the Paper*

Section 2 summarizes the dipole spin-down and pair-cascade turn-off equations and defines the canonical ATNF sample ($N = 2,356$). Section 3 derives the weak-form 2D continuity inversion, specifies the discrete basis, quadrature, and two-stage Cholesky NNLS solver, analyzes the singular spectrum and resolution matrix $\mathbf{R}$, introduces the non-circular visibility gauge calibration, and presents the ATNF reconstruction and branch-free magnetic field decay (h_0) scan. Section 4 presents 100-seed 2D map-recovery validation and the 300-seed Monte Carlo dose-response and modified frequentist CL_s calibration across Protocols P1–P4 and a parametric Gaussian baseline. Section 5 reports out-of-sample 50/50 cross-validation, cross-survey latitude transfer, Galactic core-collapse supernova birthrate concordance, and blind benchmarking on external 3D FGK06 synthetic catalogues. Section 6 evaluates four gauge-free observational cross-checks (Tests 1–4). Section 7 synthesizes the methodological and astrophysical conclusions.

2. PHYSICAL FORWARD MODEL AND OBSERVATIONAL SAMPLE

2.1. *Magnetic Dipole Spin-Down and Chronological Age*

For a neutron star with canonical moment of inertia $I \approx 10^{45}$ g cm^2, radius $R \approx 10^6$ cm, spin period P, and effective equatorial surface dipole field B, equating the rotational energy loss rate $\dot{E} = 4\pi^2 I\dot{P}/P^3$ to the magnetospheric dipole braking power $P_{\rm dip} = 32\pi^4 R^6 B^2/(3c^3P^4)$ (Ostriker & Gunn 1969; Goldreich & Julian 1969; Spitkovsky 2006; Philippov et al. 2014) yields the canonical dipole spin-down law:

$$P\dot{P} = K_0 B^2 \quad \Longleftrightarrow \quad \dot{P}(P,B) = \frac{K_0 B^2}{P}, \qquad K_0 \equiv \frac{8\pi^2 R^6}{3Ic^3} \approx 9.766 \times 10^{-40}\ \mathrm{s\ G^{-2}}, \tag{4}$$

corresponding to $B = 3.2 \times 10^{19}\sqrt{(P/\mathrm{s})\,\dot{P}}$ G (Manchester et al. 2005). Under constant B and dipole braking index $n = 3$ ($\dot{P} \propto P^{2-n}$; Espinoza et al. 2011), integrating Equation (4) from birth ($t = 0, P = P_0$) to the present (t, P) relates the true chronological age t to the observable characteristic age τ_c:

$$t = \frac{P^2 - P_0^2}{2K_0B^2} = \tau_c\left[1 - \left(\frac{P_0}{P}\right)^2\right], \tag{5}$$

where

$$\tau_c \equiv \frac{P}{2\dot{P}} = \frac{P^2}{2K_0B^2} \approx 15.8\ \mathrm{Myr}\left(\frac{P}{\mathrm{s}}\right)^2\left(\frac{B}{10^{12}\ \mathrm{G}}\right)^{-2}. \tag{6}$$

When $P_0 \ll P$, $t \to \tau_c$, whereas a pulsar born at a slow period $P_0 \in [0.40\ \mathrm{s}, P]$ inside $P \in [0.40, 0.85]$ s has a true chronological age $t < 0.779\,\tau_c$. Moreover, because $\tau_c \propto P^2/B^2$, the time required to reach $P = 0.50$ s ranges from ~ 0.10 Myr at $B = 10^{12.8}$ G to ~ 99 Myr at $B = 10^{11.3}$ G.

2.2. *Pair-Cascade Turn-Off and the Radio Death Valley*

Coherent radio emission requires electron–positron ($e^{\pm}$) pair cascades across the polar-cap potential drop $\Delta\Phi_{\rm max} = 2\pi^2R^3B/(c^2P^2) \propto B/P^2$ (Sturrock 1971; Ruderman & Sutherland 1975). Quenching of pair cascades below a threshold voltage $\Phi_{\rm crit} \sim 10^{12}$ V defines a linear radio death line in the $(x, y) \equiv (\log_{10} P, \log_{10} B)$ plane (Chen & Ruderman 1993):

$$\log_{10}\left(\frac{B/\mathrm{G}}{(P/\mathrm{s})^2}\right) = \log_{10} B - 2\log_{10} P = d_{\rm crit}, \tag{7}$$

with canonical intercept $d_{\rm crit} = 11.23$ and logistic Death Valley width $w_{\rm death} = 0.38$ dex arising from multipolar radius-of-curvature and inclination-angle variations. Solving Equation (7) for the turn-off period at field strength B gives

$$P_{\rm death}(B) = \left(\frac{B}{10^{d_{\rm crit}}\ \mathrm{G}}\right)^{1/2}\ \mathrm{s}. \tag{8}$$

Evaluated at the upper edge of the slow-birth test window ($P_2 = 0.85$ s, $\log_{10} P_2 = -0.0706$), Equation (7) yields a minimum death-line magnetic field of $\log_{10} B_{\rm min}(0.85\ \mathrm{s}) = 11.23 + 2(-0.0706) = 11.09$ ($B_{\rm min} = 1.23 \times 10^{11}$ G). Consequently, every magnetic field stratum with $\log_{10} B \geq 11.15$ ($P_{\rm death} \geq 0.91$ s $>$ 0.85 s) traverses the entire $[0.40, 0.85]$ s test window prior to pair-cascade turn-off.

2.3. *ATNF v2.7.0 Sample Selection and Weighting Construction*

Starting from all $N_{\rm raw} = 4,345$ entries in the ATNF Pulsar Catalogue (v2.7.0; Manchester et al. 2005), we isolate canonical rotation-powered Galactic disk pulsars via four reproducible cuts: (i) valid timing solutions ($P > 0$ and $\dot{P} > 0$, removing non-spinning or accretion-powered/cluster-accelerated $\dot{P} \leq 0$ sources; 3,572 remain), (ii) exclusion of binary systems ('BINARY' is null) and globular cluster or Magellanic Cloud associations ('ASSOC' excluding 'GC', 'SMC', 'LMC'; 3,088 remain), (iii) exclusion of recycled millisecond pulsars ($P < 0.03$ s or $\log_{10} B < 10.5$; 2,418 remain), and (iv) active phase-space domain restriction ($-1.5 < \log_{10} P < 0.8$ and $\log_{10} B < 13.4$, removing ultra-high-B magnetars $B \geq 10^{13.4}$ G), yielding the canonical sample of $N = 2,356$ isolated radio pulsars (Figure 1a). Within this sample, pulsars flagged as Parkes Multibeam Survey ('pksmb') members with $|b| \leq 5°$ number $N = 971$; the

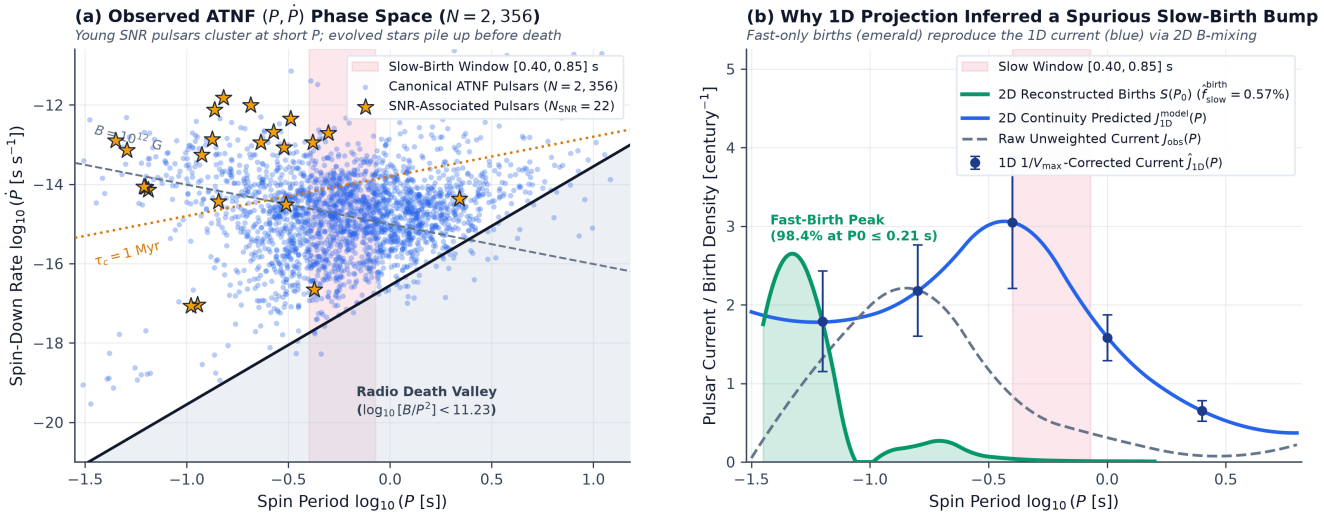


Figure 1. The ATNF $(P, \dot{P})$ phase space and the 1D current projection artifact. (a) Spin-down rate $\log_{10}\dot{P}$ versus spin period $\log_{10}P$ for $N = 2,356$ canonical isolated Galactic radio pulsars (blue points) and $N_{\rm SNR} = 22$ young pulsars in supernova remnants (gold stars). Young SNR pulsars cluster at $P < 0.40$ s, whereas the field population accumulates at $P \sim 0.3$–1.0 s above the pair-production radio death line $\log_{10}(B/P^2) = 11.23$ (solid black line; Death Valley shaded gray). **(b)** One-dimensional pulsar current $J(P) = \dot{P}\,n(P)$ (navy points, $1/V_{\rm max}$ weighted) and the 2D reconstructed birth period distribution $S(P_0)$ (shaded emerald curve, $\hat{f}^{\rm birth}_{\rm slow} = 0.57\%$). Collapsing the 2D (P, B) flow onto 1D period bins produces a pronounced apparent current peak near $P \approx 0.40$ s (matched by our 2D continuity pushforward, solid blue curve) even when $> 98\%$ of births occur at $P_0 < 0.40$ s.

broader $|b| \leq 5°$ plane footprint (including non-`pksmb` surveys) contains $N = 1,039$ pulsars. The intermediate-distance annulus $D \in [1, 4]$ kpc contains $N = 944$ pulsars. Throughout this paper, "`pksmb`" refers to the $N = 971$ survey-flagged subsample.

For 1D current diagnostics (Figure 1b and Section 6.3), we define a baseline inverse-volume/beaming weight $w_k \propto [V_{\rm max}(L_{1400,k})\, f_{\rm beam}(P_k)]^{-1}$ truncated at $w_k \leq 50\,{\rm median}(w)$. Across the $N = 2,356$ sample, 1,556 pulsars (66.0%) have directly measured 1400 MHz flux densities S_{1400} and YMW16 dispersion-measure distances D (Yao et al. 2017) giving $L_{1400} = S_{1400}D^2$, whereas 800 pulsars (34.0%) lack published S_{1400} values; for those 800 objects, w_k is assigned via a Gaussian Nadaraya–Watson conditional kernel lookup $E[w \mid x, y]$ with bandwidths $(h_x, h_y) = (0.12, 0.15)$ dex. Because imputed flux weights can transfer selection trends across surveys (Section 5.1), our primary kinematic clock (Section 6.2) and strip conservation bound (Section 6.4) use strictly unweighted ($w_k = 1$) counts, and in Section 4 we verify that replacing catalog weights with uniform weights ($w_k = 1$) in the 2D PDE inversion leaves the reconstructed slow-birth fraction unchanged ($\hat{f}^{\rm birth}_{\rm slow} \leq 1\%$).

3. THE WEAK-FORM 2D CONTINUITY INVERSION METHOD

3.1. *Continuous 2D Transport PDE and Weak-Form Variational Formulation*

Let $f_{\rm true}(P, B)$ denote the steady-state Galactic phase-space number density of active pulsars per unit spin period P and magnetic field B. Under secular period evolution $\dot{P}(P, B) = K_0 B^2/P$, exponential magnetic field evolution $\dot{B}(B) = -h_0 B$ (with decay rate $h_0 \geq 0$ in Myr^{-1}, corresponding to Ohmic decay timescale $\tau_{\rm Ohm} = 1/h_0$), effective mortality/turn-off rate $\kappa + D_{\rm death}(P, B)$, and birth source density $S_{\rm true}(P, B) \geq 0$, stationary phase-space conservation ($\partial_t f_{\rm true} = 0$) obeys the 2D advection–reaction partial differential equation:

$$\partial_P[\dot{P} f_{\rm true}] + \partial_B[\dot{B} f_{\rm true}] + [\kappa + D_{\rm death}(P, B)] f_{\rm true} = S_{\rm true}(P, B). \quad (9)$$

Because a catalogue of $N = 2,356$ discrete pulsars defines an empirical measure $f_{\rm obs}(P, B) = \sum_{k=1}^{N} W_k\, \delta(P - P_k)\, \delta(B - B_k)$ (with normalized weights $W_k = w_k / \sum_j w_j$), evaluating the partial derivatives ∂_P and ∂_B directly on histogram bins amplifies Poisson shot noise in proportion to $(\Delta P)^{-1}$ and $(\Delta B)^{-1}$.

To eliminate numerical differentiation of the data, we write $f_{\rm true}(P, B) = f_{\rm obs}(P, B)/\beta_{\rm vis}(P, B)$ in terms of the dimensionless relative visibility function $\beta_{\rm vis}(P, B) \equiv \exp[G_{\rm prim}(x, y)]$, multiply Equation (9) by $\beta_{\rm vis}(P, B)$ and a smooth, compactly supported C^2 test function $\psi_m(x, y)$ in logarithmic coordinates $(x, y) = (\log_{10} P, \log_{10} B)$, and integrate by parts over phase space. Because $\psi_m(x, y)$ vanishes on the active boundary prior to the Death Valley cut, all boundary terms

vanish identically, yielding the exact *weak-form continuity identity*:

$$\begin{aligned}&\iint \psi_m(x,y)\,\beta_{\rm vis}(P,B)\,S_{\rm true}(P,B)\,dP\,dB\\&=\sum_{k=1}^{N} W_k\Bigg[-\dot{P}_k\frac{\partial_x\psi_m(x_k,y_k)}{P_k\ln 10}+h_0B_k\frac{\partial_y\psi_m(x_k,y_k)}{B_k\ln 10}\\&\qquad+\left(\kappa-\frac{dG_{\rm prim}}{dt}\Big|_k\right)\psi_m(x_k,y_k)\Bigg],\end{aligned}\tag{10}$$

where the convective derivative of the log-visibility gauge along the characteristic trajectory is

$$\frac{dG_{\rm prim}}{dt}=\frac{\partial G_{\rm prim}}{\partial x}\frac{\dot{P}}{P\ln 10}-\frac{\partial G_{\rm prim}}{\partial y}\frac{h_0}{\ln 10}.\tag{11}$$

We pair the differential flux moments of Equation (10) with integral Green's pushforward density moments obtained by integrating Equation (9) along analytical characteristic trajectories $(P(t;P_0,B_0),B(t;B_0))$ from birth ($t=0$) to death ($t=t_{\rm max}$):

$$\begin{aligned}&\iint S_{\rm true}(P_0,B_0)\,dP_0\,dB_0\int_0^{t_{\rm max}(P_0,B_0)}\psi_m^{\rm dens}(x(t),y(t))\\&\times\chi_{\rm death}(x(t),y(t))\,\beta_{\rm vis}(P(t),B(t))\,e^{-\kappa t}\,dt\\&=\sum_{k=1}^{N}W_k\,\psi_m^{\rm dens}(x_k,y_k),\end{aligned}\tag{12}$$

where $\chi_{\rm death}(x,y)=\{1+\exp[-4(y-2x-d_{\rm crit})/w_{\rm death}]\}^{-1}$ is the smooth logistic survival factor across Death Valley. Equations (10) and (12) evaluate derivatives exclusively on the analytic test functions ψ_m and $\beta_{\rm vis}$, reducing the empirical right-hand side to exact Monte Carlo sums over the observed pulsars.

3.2. *Discrete Source Grid, Test-Function Families, Quadrature, and Two-Stage NNLS Solver*

We discretize the birth source density $S(x_0,y_0)$ on a $N_{X_S}\times N_{Y_S}=14\times 15=210$ tensor-product grid spanning $x_0\in[-1.45,0.20]$ ($P_0\in[0.035,1.585]$ s, spacing $\Delta x_0=0.1269$ dex) and $y_0\in[10.65,13.40]$ ($B_0\in[4.47\times10^{10},2.51\times10^{13}]$ G, spacing $\Delta y_0=0.1964$ dex). Each source cell (i,j) is represented by a biquartic kernel of half-widths $h_{x,S}=1.15\,\Delta x_0$ and $h_{y,S}=1.15\,\Delta y_0$; cells lying below the death-line buffer $y_0-2x_0<d_{\rm crit}+0.35$ are zeroed identically. Within each active source cell, birth integrals are evaluated using a $N_{UX}\times N_{UY}=7\times7$ tensor-product quadrature rule with quartic weights $w_u=(1-u^2)^2$, and the characteristic streamline integral in Equation (12) is evaluated using $N_T=48$ quadratic time nodes $t_k=t_{\rm max}s_k^2$ ($s_k=(k-1/2)/N_T$, $dt_k=2t_{\rm max}s_k/N_T$) along the exact analytical solution of the dipole-plus-decay trajectory:

$$\begin{aligned}P(t;P_0,B_0)&=\sqrt{P_0^2+2K_0B_0^2\,t\,\mathrm{exprel}(-2h_0t)},\\B(t;B_0)&=B_0e^{-h_0t},\qquad \mathrm{exprel}(z)\equiv\frac{e^z-1}{z},\end{aligned}\tag{13}$$

where $\mathrm{exprel}(-2h_0t)$ is evaluated via its Taylor series for $|2h_0t|<10^{-4}$ so that the operator is C^∞-smooth down to $h_0=0$. The upper integration limit $t_{\rm max}(P_0,B_0)=\min(t_{\rm death},12/\kappa,450\ \mathrm{Myr})$ is computed from the exact Citardauq quadratic root of $\log_{10}(B(t)/P(t)^2)=d_{\rm crit}-1.2\,w_{\rm death}$.

The test-function bank comprises two families built from the 1D C^2 quartic bump $B_1(u)=(1-u^2)^2\,\Theta(1-|u|)$ and the 1D C^1 monotone step-down function $S_1(u)=(\frac{1}{2}-\frac{3}{4}u+\frac{1}{4}u^3)\,\Theta(1-|u|)+\Theta(-1-u)$:

1. *Density test cells ($M_{\rm dens}=390$)*: A 15×13 grid of centers $(x_c,y_c)\in[-1.35,0.45]\times[10.65,13.30]$, where each center contributes both a cumulative step-down cell $S_1((x-x_c)/0.14)\,B_1((y-y_c)/0.22)$ and a localized bump cell $B_1((x-x_c)/0.20)\,B_1((y-y_c)/0.22)$.

2. *Differential flux test cells ($M_{\rm diff}=123$)*: A 15×12 grid of step-down cells ($w_x=0.14, w_y=0.22$) on $[-1.35,0.25]\times[10.90,13.30]$ plus a 10×10 grid of bump cells ($w_x=0.22, w_y=0.22$) on $[-0.55,0.40]\times[10.90,13.30]$, restricted strictly to active phase space satisfying $(y_c-0.22)-2(x_c+w_x)\geq d_{\rm crit}+0.15$.

Each of the $M_{\rm tot}=M_{\rm dens}+M_{\rm diff}=513$ moment equations is studentized by its fixed-reference Campbell shot-noise standard deviation $\sigma_m=\sqrt{\sum_{k=1}^{N}W_k^2g_{m,k}^2+\sigma_{{\rm floor},m}^2}$, with variance floors $\sigma_{\rm floor,dens}=0.002$ and $\sigma_{\rm floor,diff}(y_c)=\max[0.003\times10^{1.5(y_c-12)},0.0025]$ held invariant across parameter scans so the χ^2 metric does not shift with $\boldsymbol{\theta}$.

To factor out the B_0^{-2} variation in pulsar lifetimes across the 15 magnetic-field columns, we reparameterize the unknown source vector in terms of the residence-time-normalized birth weights $n_{ij}\equiv S_{ij}\,\bar{\tau}_j\geq0$, where $\bar{\tau}_j=\max(W_{1,j}^{\rm raw},10^{-12})$ is the streamline residence norm of column j. Balancing the Frobenius norms of the differential and density blocks defines the joint design matrix $\tilde{\mathbf{A}}\in\mathbb{R}^{513\times210}$ and right-hand side $\tilde{\mathbf{b}}\in\mathbb{R}^{513}$. We solve for $\mathbf{n}\in\mathbb{R}^{210}_{\geq0}$ via a two-stage active-set Lawson–Hanson Non-Negative Least Squares (NNLS) algorithm:

$$\begin{aligned}\hat{\mathbf{n}}&=\arg\min_{\mathbf{n}\geq\mathbf{0}}\left\{\|\tilde{\mathbf{A}}\mathbf{n}-\tilde{\mathbf{b}}\|_2^2+\mathbf{n}^T\mathbf{R}_{\rm pen}\mathbf{n}\right\},\\\mathbf{R}_{\rm pen}&=\left(\lambda_x\mathbf{D}_{2,x}^T\mathbf{D}_{2,x}+\lambda_y\mathbf{D}_{2,y}^T\mathbf{D}_{2,y}\right)\odot(\mathbf{s}\mathbf{s}^T),\end{aligned}\tag{14}$$

where $\mathbf{D}_{2,x}$ and $\mathbf{D}_{2,y}$ are second-difference operators along x_0 and y_0 ($\lambda_x=2.0\times10^{-3}$, $\lambda_y=0.35$, with $\mathbf{D}_{2,y}$ decoupled above $y_0>13.05$). In Stage 1, a

preliminary isotropic ridge solve ($\mathbf{R}_1 = 2 \times 10^{-3}\bar{s}^2\mathbf{I}$) estimates the marginal B_0 visibility weights $\hat{w}_{y,j} = \max(m_j/\sum_{j'} m_{j'}, 0.006)$; in Stage 2, setting $s_{ij} = \bar{s}/(6.0\,\hat{w}_{y,j})$ equalizes the smoothing penalty across low-count and high-count magnetic-field strata, and the Cholesky factor $\mathbf{L}$ of $\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_{\rm pen} = \mathbf{L}\mathbf{L}^T$ reduces Equation (14) to the standard positive-definite NNLS problem $\min_{\mathbf{n}\geq\mathbf{0}} \|\mathbf{L}^T\mathbf{n} - \mathbf{L}^{-1}\tilde{\mathbf{A}}^T\tilde{\mathbf{b}}\|_2^2$.

3.3. *Condition Number, Singular Spectrum, and Effective Degrees of Freedom*

Evaluating the forward operator at the canonical ATNF baseline ($h_0 = 0.015$ Myr^{-1}, $\kappa = 0.005$ Myr^{-1}, $d_{\rm crit} = 11.23$, $w_{\rm death} = 0.38$) on the $N_{\rm active} = 191$ active source cells (19 of the 210 cells lie below the deathline buffer and are identically zero) demonstrates why residence-time preconditioning and Tikhonov regularization are essential:

1. *Conditioning across preconditioning stages*: Because dipole spin-down lifetimes scale as B_0^{-2} across 2.75 decades of B_0, the unpreconditioned operator $\mathbf{A}_{\rm unscaled}$ acting on raw S_{ij} has singular values spanning 7.70×10^2 down to 1.41×10^{-3}, giving an ill-conditioned 2-norm condition number $\kappa_2(\mathbf{A}_{\rm unscaled}) = 5.45 \times 10^5$. Dividing by the column residence norms $\bar{\tau}_j$ reduces the condition number by a factor of 18 to $\kappa_2(\tilde{\mathbf{A}}) = 3.07 \times 10^4$ ($\sigma_{\max} = 5.565$, $\sigma_{\min} = 1.815 \times 10^{-4}$). Including the Tikhonov penalty yields a well-conditioned symmetric positive-definite normal matrix with $\kappa_2(\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_1) = 2.24 \times 10^3$ in Stage 1 and $\kappa_2(\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_{\rm pen}) = 2.41 \times 10^5$ in Stage 2, more than ten orders of magnitude below the IEEE-754 double-precision limit ($\epsilon_{\rm mach}^{-1} \approx 4.5 \times 10^{15}$).

2. *Effective degrees of freedom*: The linear hat (influence) matrix of the regularized operator is $\mathbf{H} = \tilde{\mathbf{A}}(\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_{\rm pen})^{-1}\tilde{\mathbf{A}}^T$. Its trace gives the effective number of fitted source parameters, $\mathrm{df}_{\rm model} = \mathrm{Tr}(\mathbf{H}) = 21.3$ (out of 210 grid nodes), leaving $\nu_{\rm eff} = M_{\rm tot} - \mathrm{Tr}(\mathbf{H}) = 513 - 21.3 = 491.7$ effective residual degrees of freedom.

3.4. *Resolution Matrix* $\mathbf{R}$*, Averaging Kernels, and Analytical Origin of the Dose-Response Slope*

In geophysical and astrostatistical inverse theory (Backus & Gilbert 1968; Tarantola 2005), the linear mapping between the true underlying source distribution $\mathbf{n}_{\rm true}$ and its regularized expectation $\mathbb{E}[\hat{\mathbf{n}}] = \mathbf{R}\,\mathbf{n}_{\rm true}$ is governed by the 210×210 *model resolution matrix*:

$$\mathbf{R} \equiv \left(\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_{\rm pen}\right)^{-1}\tilde{\mathbf{A}}^T\tilde{\mathbf{A}}, \qquad \mathrm{Tr}(\mathbf{R}) = \mathrm{Tr}(\mathbf{H}) = 21.3. \tag{15}$$

Row k of $\mathbf{R}$ is the 2D *averaging kernel* (point-spread function) showing how a unit point source at grid cell k is smoothed across the 14×15 birth grid by the regularized inversion.

Figure 2b displays the peak-normalized 1D averaging kernels along $\log_{10} P_0$ at representative fast-birth ($x_0 = -1.07$, $P_0 = 0.085$ s), left-boundary ($x_0 = -1.45$, $P_0 = 0.035$ s), and slow-window ($x_0 = -0.18$, $P_0 = 0.66$ s) nodes for $\log_{10} B_0 = 12.22$. Across all active interior cells, the median full width at half maximum is $\mathrm{FWHM}(\log_{10} P_0) = 0.475$ dex (interquartile range $[0.348, 0.619]$ dex; $\mathrm{FWHM}_x = 0.293$ dex inside the slow window at $x_0 = -0.18$, 0.208 dex at the left boundary $x_0 = -1.45$, and 0.617 dex in the fast-spin interior at $x_0 = -1.07$) and $\mathrm{FWHM}(\log_{10} B_0) = 0.577$ dex (IQR $[0.474, 0.740]$ dex). Thus, the 2D continuity operator resolves variations in initial magnetic field B_0 to ~ 0.58 dex and variations in initial spin period P_0 inside the slow-birth window to ~ 0.29 dex (≈ 2.3 grid cells), while broader ~ 0.62 dex averaging occurs in the rapid-spin interior ($P_0 \lesssim 0.15$ s) where initial period differences $(P_0/P)^2 \ll 1$ wash out along spin-down trajectories (Equation 5).

Crucially, the resolution matrix $\mathbf{R}$ provides a closed-form analytical explanation for the empirical dose-response slope in Section 4: applying $\mathbf{R}$ directly to our slow-birth injection profile centered at $x_0 = -0.20$ ($P_0 = 0.63$ s, $\sigma_x = 0.08$ dex) retains exactly 76.9% of the injected slow-birth probability mass inside the target window $P_0 \in [0.40, 0.85]$ s while smoothing 23.1% across the window boundaries at 0.40 s and 0.85 s (accounting for the ≈ 0.77 slope).

3.5. *Visibility Gauge Parameterization, Non-Circular Calibration, and Window Boundary Sensitivity*

The relative detection probability $\beta_{\rm vis}(P, B) = \exp[G_{\rm prim}(x, y) - c_1 \ln 10\,(-1.0)]$ is parameterized by a three-parameter, C^1-smooth hinged log-quadratic gauge:

$$G_{\rm prim}(x, y) = \ln 10\Big[c_1 x + \tfrac{1}{2}c_2\,\max(x - x_{\rm knee}, 0)^2 + c_3\,\phi_0(x)\,(y - 12.0)\Big], \tag{16}$$

where $x_{\rm knee} = -0.65$ ($P_{\rm knee} = 0.224$ s) and $\phi_0(x)$ is the C^1 quadratic-to-linear ramp that vanishes for $x \leq -0.65$, equals $2(x + 0.65)^2$ on $-0.65 < x < -0.40$, and equals $(x + 0.65) - 0.125$ for $x \geq -0.40$. Each term in Equation (16) has a direct physical origin: $c_1 \approx -0.25$ captures the geometric sky-solid-angle beaming law $f_{\rm beam}(P) \propto P^{-1/3}$ ($d\log_{10} f_{\rm beam}/dx \approx -0.33$; Lyne & Manchester 1988; Tauris & Manchester 1998) at short periods; $c_2 < 0$ captures the steepening radio luminosity drop-off as spin-down power $\dot{E} \propto P^{-3}$ falls at $P > P_{\rm knee}$; and $c_3 > 0$ captures the magnetic-field dependence of polar-cap pair-cascade efficiency ($\dot{E} \propto B^2/P^3$) at long periods.

To guarantee that the gauge parameters (c_1, c_2, c_3) cannot absorb a true slow-birth signal inside the test

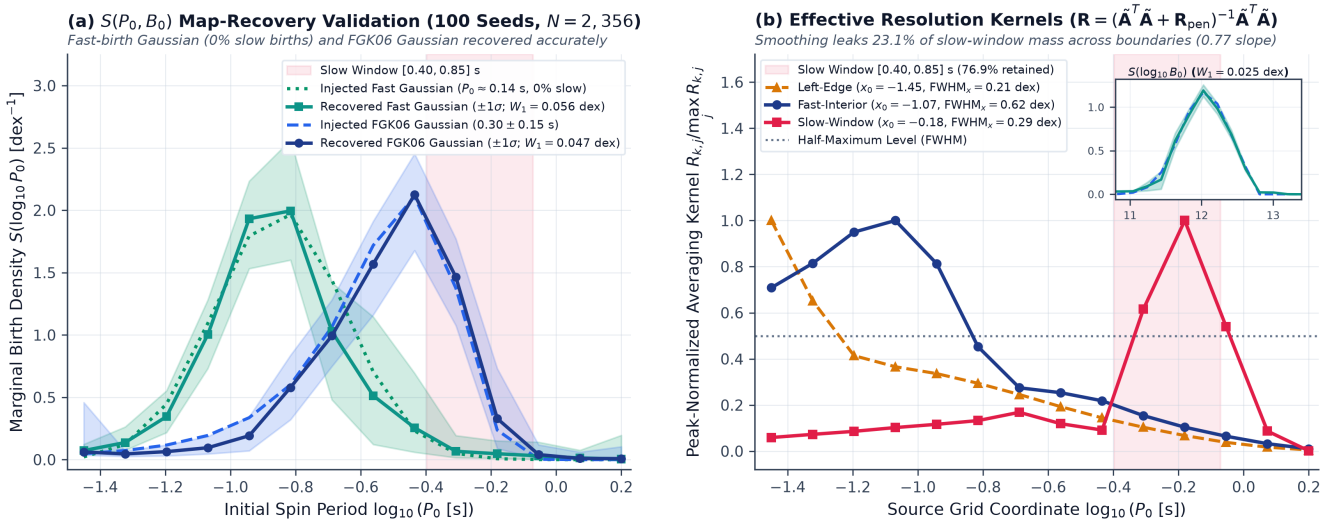


Figure 2. Synthetic $S(P_0, B_0)$ map-recovery validation and resolution averaging kernels. (a) Marginal birth period density $S(\log_{10} P_0)$ recovered across 100 synthetic Monte Carlo catalogues ($N = 2,356$) drawn from two distinct ground truths: a unimodal fast-birth Gaussian ($\log_{10} P_0 \sim \mathcal{N}(-0.85, 0.20^2)$, $P_0 \approx 0.14$ s, 0% slow births; green dotted curve vs. teal solid median and $\pm 1\sigma$ band, ensemble-median $W_1 = 0.019$ dex) and the canonical unimodal FGK06 Gaussian ($P_0 \sim \mathcal{N}(0.30\,\text{s}, 0.15^2\,\text{s}^2)$; blue dashed curve vs. navy solid median and $\pm 1\sigma$ band, ensemble-median $W_1 = 0.034$ dex). Both Gaussians are recovered accurately across the full spin-period domain without spurious pile-up inside the $[0.40, 0.85]$ s slow window (shaded coral). **(b)** Peak-normalized rows of the resolution matrix $\mathbf{R} = (\tilde{\mathbf{A}}^T\tilde{\mathbf{A}} + \mathbf{R}_{\text{pen}})^{-1}\tilde{\mathbf{A}}^T\tilde{\mathbf{A}}$ at left-edge ($x_0 = -1.45$, FWHM$_x = 0.21$ dex), fast-interior ($x_0 = -1.07$, FWHM$_x = 0.62$ dex), and slow-window ($x_0 = -0.18$, FWHM$_x = 0.29$ dex) source nodes; smoothing across the edges of $[0.40, 0.85]$ s retains 76.9% of slow-window mass. Inset: marginal initial magnetic-field recovery $S(\log_{10} B_0)$ ($W_1 = 0.025$ dex median).

window $P \in [0.40, 0.85]$ s ($x \in [-0.398, -0.070]$), we fit (c_1, c_2, c_3) via a *non-circular Markov conditional maximum-likelihood estimator* evaluated strictly on two disjoint source-free calibration intervals outside $[0.40, 0.85]$ s: (i) a short-period pre-window strip $x \in [-0.72, -0.42]$ ($P \in [0.19, 0.38]$ s) and (ii) a late-period post-window tail $x \in [-0.05, 0.55]$ ($P \in [0.89, 3.55]$ s). Within each calibration interval, conditional on a pulsar's magnetic field y_k, its period density along the spin-down streamline is proportional to $(dP/dt)^{-1}\chi_{\text{death}}(x, y_k)\beta_{\text{vis}}(P, y_k) \propto P^2\chi_{\text{death}}(x, y_k)\exp[G_{\text{prim}}(x, y_k)]$; maximizing the conditional log-likelihood over (c_1, c_2, c_3) via L-BFGS-B requires zero knowledge of the birth distribution inside $[0.40, 0.85]$ s. We note one limitation: the pre-window calibration interval $[0.19, 0.38]$ s partially overlaps the FGK06 Gaussian birth distribution (which has $\sim$ 30% of its weight below 0.40 s), so the gauge is not strictly "source-free" under a broad unimodal birth model. Case A S-recovery (Section 4.1) shows that the solver overshoots by 2.7 pp (23.1% vs. 20.5%) in this regime, providing a direct empirical measure of this leakage.

We further verify that our conclusions are insensitive to the exact placement of the slow-birth test window $[P_1, P_2]$ by repeating both the ATNF reconstruction and 100-seed Protocol P2 Monte Carlo calibrations across four window definitions:

1. *Shifted-early window* $[0.35, 0.80]$ *s* ($x_0 \in [-0.456, -0.097]$): $\hat{f}_{\text{slow}}^{\text{birth}} = 1.85\%$ on ATNF; 95% CL$_s$ upper limit = 19.4%.
2. *Canonical window* $[0.40, 0.85]$ *s* ($x_0 \in [-0.398, -0.070]$): $\hat{f}_{\text{slow}}^{\text{birth}} = 0.57\%$ on ATNF; 95% CL$_s$ upper limit = 28% (27.5%).
3. *Shifted-late window* $[0.45, 0.90]$ *s* ($x_0 \in [-0.347, -0.046]$): $\hat{f}_{\text{slow}}^{\text{birth}} = 0.05\%$ on ATNF; 95% CL$_s$ upper limit = 21.1%.
4. *Widened window* $[0.35, 0.90]$ *s* ($x_0 \in [-0.456, -0.046]$): $\hat{f}_{\text{slow}}^{\text{birth}} = 1.85\%$ on ATNF; 95% CL$_s$ upper limit = 20.1%.

Across all four window definitions, the ATNF point estimate remains $\leq$ 1.85% and the 95% CL$_s$ exclusion limits remain within 19.4%–27.5%, confirming that the exclusion of classical injection does not depend on the exact window boundaries. The somewhat weaker limit for the canonical window reflects its alignment with the source grid: the 0.328 dex window spans ≈ 2.6 cells of width $\Delta x_0 = 0.127$ dex, so the placement of cell boundaries relative to the window edges modestly affects the recovered fraction.

3.6. *ATNF Reconstruction, Bootstrap Uncertainty Bands, and Magnetic Field Decay (h_0) Scan*

Figure 3a displays the reconstructed 2D birth density $S(\log_{10} P_0, \log_{10} B_0)$ and its 1D marginal $S(\log_{10} P_0)$ with 68% and 95% nonparametric bootstrap uncertainty bands (200 catalogue resamples with per-resample Markov gauge refitting) at the canonical baseline $h_0 = 0.015\ \mathrm{Myr}^{-1}$. The birth distribution is concentrated at short initial spin periods, peaking in the interior at $\log_{10} P_0 = -1.32$ ($P_0 \approx 48$ ms) with 94.46% of all reconstructed births occurring at $P_0 \leq 0.21$ s (bootstrap mean $92.27\% \pm 7.60\%$, 95% CI [73.20%, 99.90%]), while the left-boundary cell at $x_0 = -1.45$ ($P_0 = 35$ ms) accounts for 26.0% of the marginal weight (Table 2). The slow-birth fraction in $P_0 \in [0.40, 0.85]$ s is $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}} = 0.57\%$ in n_w (1.24% in S_w; bootstrap median 1.03%, 95% CI [0.10%, 7.07%]).

To test whether the inferred absence of slow births depends on the assumed magnetic field decay rate h_0 or on low-B Death Valley modeling, we evaluate the inversion across 10 values of $h_0 \in [0, 0.048]\ \mathrm{Myr}^{-1}$ ($\tau_{\mathrm{Ohm}} \in [20.8\ \mathrm{Myr}, \infty)$) under three gauge prescriptions—interior-profiled (κ, c_1, c_2, c_3), noncircular Markov conditional MLE, and a zero-free-parameter physical $S_{1400}(\dot{E})$ Malmquist gauge—and repeat the scan on the masked subsample with $\log_{10} B \geq 11.15$ ($N = 2,068$), where no pulsar dies inside [0.40, 0.85] s (Figure 3b and Table 1). Across the entire scan, $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}}$ remains bounded within [0.83%, 9.03%] on the full catalogue and drops below 1.63% for $h_0 \geq 0.03\ \mathrm{Myr}^{-1}$, confirming the absence of large slow injection regardless of field decay. We adopt the conservative upper envelope $\hat{f}^{\mathrm{birth}}_{\mathrm{obs,max}} \leq 9.03\%$ across unconstrained h_0 as the full parametric bound.

To evaluate whether the inferred fast-birth mode and absence of slow injection are sensitive to domain boundaries or observational period cuts, we perform a two-dimensional grid scan over the domain lower edge $\log_{10} P_{\mathrm{min}} \in \{-1.75, -1.55, -1.35\}$ (17.8–44.7 ms) and sample selection cut $P_{\mathrm{cut}} \in \{0.02, 0.03, 0.05\}$ s using consistent survey-selection weights (Table 2). Across all 9 configurations the slow-birth result is stable: $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}} \leq 1.09\%$ in n_w ($\leq 1.80\%$ in S_w), at least 92.9% of births lie at $P_0 \leq 0.21$ s, and the reconstructed peak lies in $P_0 \in [43.1, 59.8]$ ms. Evaluating intermediate period cuts on a fine grid $P_{\mathrm{cut}} \in \{0.020, 0.025, 0.030, 0.035, 0.040, 0.045, 0.050\}$ s at the canonical domain edge ($\log_{10} P_{\mathrm{min}} = -1.55$) gives the same peak ($P_0 = 47.5$ ms for all 7 cuts), the same gauge parameters $(c_1, c_2, c_3) = (-0.239, -2.493, 0.486)$, $\chi^2 \in [235.8, 246.6]$, and $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}} = 0.57\% \pm 0.01\%$. The invariance of the gauge parameters follows from the calibration windows ($P \in [0.19, 0.38]$ s and $P \in [0.89, 3.55]$ s), which lie entirely above the range of the period cuts. The position of the fast peak is resolved only to within the discretization of the lower domain edge: moving the edge to −1.35 (44.7 ms) places the first cell center at 56.2 ms, where it absorbs 79.4%–83.3% of the marginal weight, so the fast distribution is unresolved at that boundary and is consistent with extending below 50 ms. The robustness of the slow-birth result to the gauge calibration windows, the high-period end of the sample and the $\log_{10} B$ cuts is examined in Table 3 and Figure 4.

Robustness to gauge windows, sample ceiling and field cuts. Table 3 and Figure 4 repeat the inversion of the real ATNF sample under alternative configurations. Five alternative pre-window calibration strips leave the result unchanged (0.56%–0.57% in n_w, 1.23%–1.24% in S_w; $c_2 = -2.48$ to -2.49). Moving the start of the post-window tail window to $x_{\mathrm{cut}} = 0.00$, 0.10, 0.20 and 0.30 ($P \geq 1.00$–2.00 s) changes $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}}$ by up to +0.33 pp (0.51%–0.90% in n_w, 0.92%–1.53% in S_w); c_2 moves between −2.28 and −2.80, and the $x_{\mathrm{cut}} = +0.10$ case lowers the fraction of births below 0.21 s to 87.8%, so the gauge fit shows mild sensitivity to where the tail window starts. Truncating the sample at $P \leq 3.55$, 2.51 and 2.00 s gives 0.57%, 0.57% and 0.42% (n_w). Raising the lower $\log_{10} B$ cut to 10.8, 11.0, 11.15 and 11.5 gives 0.44%–1.00% (n_w) and 1.14%–1.27% (S_w). Lowering the upper cut to 13.2, 13.0 and 12.8 *reduces* the estimate to 0.15%–0.16% (n_w) and 0.01%–0.03% (S_w), so the small nonzero canonical value is supplied by the highest-field pulsars. Across all variants $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}} \in [0.15\%, 1.00\%]$ (n_w) and [0.01%, 1.53%] (S_w), all below the 1.63% threshold of Section 4.2; the canonical value is neither the largest nor the smallest. The Markov-gauge inversion therefore does not depend materially on the calibration windows, the high-period end of the sample, or the field cuts.

4. SYNTHETIC VALIDATION, DOSE-RESPONSE, AND CL_S CALIBRATION

4.1. *2D $S(P_0, B_0)$ Map-Recovery Validation*

Before calibrating scalar upper limits on $f^{\mathrm{birth}}_{\mathrm{slow}}$, we test whether the weak-form 2D inversion accurately recovers complete 2D birth distributions $S(\log_{10} P_0, \log_{10} B_0)$ from finite catalogues of $N = 2,356$ pulsars evolved through our continuous-time 10^8 yr Markov renewal Lagrangian simulator. We evaluate 100 independent Monte Carlo realizations for each of three synthetic ground truths sharing a Log-Normal initial magnetic field distribution ($\log_{10} B_0 \sim \mathcal{N}(12.05, 0.35^2)$ truncated at 12.65):

1. *Case A (Unimodal FGK06 Gaussian, $P_0 \sim \mathcal{N}(0.30\ s, 0.15^2\ s^2)$, $f^{\mathrm{birth}}_{\mathrm{slow,inj}} = 20.5\%$ on the discrete grid)*: The solver recovers $\hat{f}^{\mathrm{birth}}_{\mathrm{slow}} = 23.1\% \pm 5.5\%$ (median 23.9%, IQR [19.9%, 27.0%]), with 1D Wasserstein distance $W_1(\log_{10} P_0) = 0.047 \pm 0.020$ dex ($W_1 = 0.034$ dex for the ensemble-median map; Figure 2a), $W_1(\log_{10} B_0) = 0.026 \pm 0.006$ dex (0.017 dex ensemble median; Figure 2b inset), and ensemble-mean 2D relative integrated

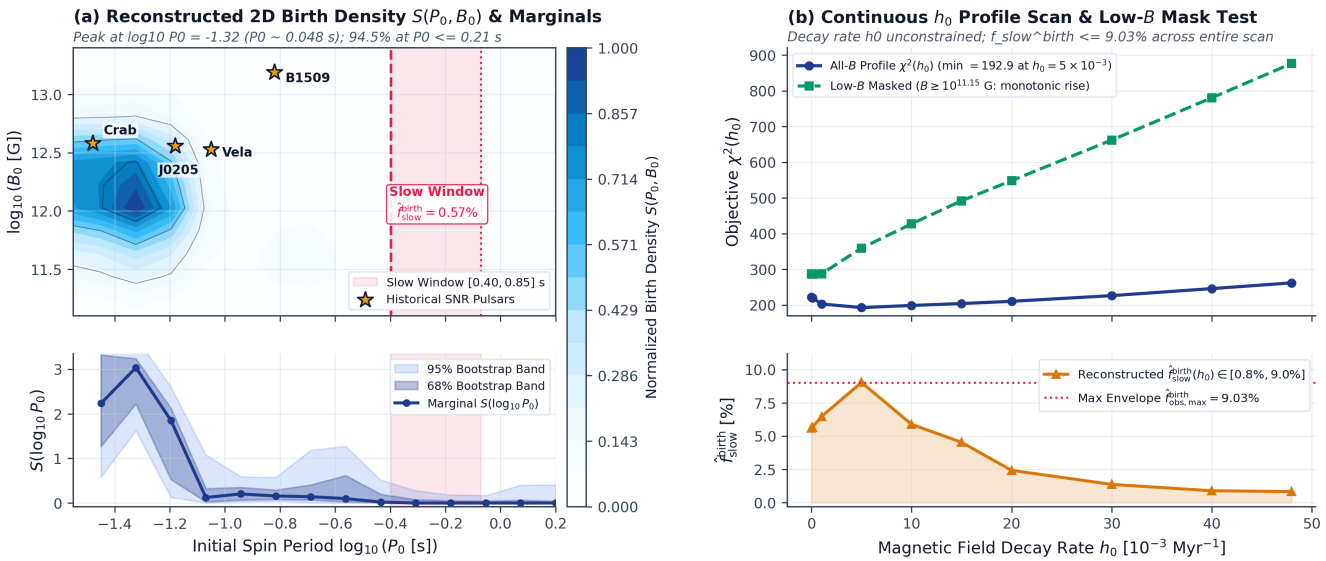


Figure 3. Reconstructed ATNF birth density $S(\log_{10} P_0, \log_{10} B_0)$, 200-bootstrap uncertainty bands, and branch-free magnetic field decay (h_0) scan. (a) Two-dimensional birth rate density $S(\log_{10} P_0, \log_{10} B_0)$ (shaded heatmap) overlaid with the 1D marginal birth period distribution $S(\log_{10} P_0)$ (white solid curve with 68% and 95% bootstrap confidence bands across 200 resamples; right axis). The 1D marginal peaks at $P_0 \approx 48$ ms (94.5% of births at $P_0 \leq 0.21$ s) and drops to $\hat{f}^{\rm birth}_{\rm slow} = 0.57\%$ (95% bootstrap CI $[0.10\%, 7.07\%]$) inside $P_0 \in [0.40, 0.85]$ s. **(b)** Profiled goodness-of-fit $\chi^2_{\rm tot}(h_0)$ (left axis) and slow-birth fraction $\hat{f}^{\rm birth}_{\rm slow}(h_0)$ (right axis) across $h_0 \in [0, 0.048]$ Myr^{-1} on the full catalogue ($N = 2,356$) and on the $B \geq 10^{11.15}$ G death-line-free subsample ($N = 2,068$). While the all-B profile has a shallow dip at $h_0 = 0.005$ Myr^{-1}, the masked low-B profile rises monotonically, proving that h_0 is unconstrained by 2D continuity, while $\hat{f}^{\rm birth}_{\rm slow} \leq 9.03\%$ across the entire scan.

Table 1. Branch-Free Interior-Profiled Magnetic Field Decay Scan ($h_0 \in [0, 0.048]$ Myr^{-1}) and Death-Line-Free Masking ($B \geq 10^{11.15}$ G)

h_0 [Myr^{-1}]	$\tau_{\rm Ohm}$ [Myr]	$\chi^2_{\rm tot}$ (Full)	$\Delta\chi^2_{\rm prof}$ (vs. Min)	$\hat{f}^{\rm birth}_{\rm slow}$ (Prof.) [%]	$\hat{f}^{\rm birth}_{\rm slow}$ (Markov) [%]	$\hat{f}^{\rm birth}_{\rm slow}$ (Phys. S_{1400}) [%]	$\chi^2_{\rm mask}$ ($B \geq 10^{11.15}$)	$\hat{f}^{\rm birth}_{\rm slow,mask}$ [%]	$D_{\rm KS}(P)/D_{\rm KS}(B)$ (Forward 10^8 yr)
0.0000	∞	222.00	+29.13	5.59%	1.63%	1.04%	287.27	7.10%	0.0431/0.0355
0.0001	10,000	219.96	+27.09	5.66%	1.57%	0.99%	286.16	7.25%	0.0426/0.0399
0.0010	1,000	202.93	+10.06	6.48%	1.20%	0.70%	287.48	8.87%	0.0442/0.0372
0.0050	200.0	192.87	0.00	9.03%	0.89%	0.47%	359.48	11.93%	0.0427/0.0363
0.0100	100.0	198.81	+5.94	5.88%	1.04%	1.34%	427.42	6.14%	0.0384/0.0322
0.0150	66.7	204.15	+11.28	4.53%	0.95%	1.33%	492.03	4.13%	0.0400/0.0333
0.0200	50.0	210.39	+17.52	2.43%	0.88%	1.47%	548.36	1.50%	0.0401/0.0334
0.0300	33.3	226.34	+33.47	1.37%	0.75%	4.50%	662.00	0.53%	0.0388/0.0311
0.0400	25.0	245.98	+53.11	0.89%	0.68%	11.35%	780.24	0.01%	0.0332/0.0312
0.0480	20.8	262.06	+69.19	0.83%	0.70%	19.95%	876.53	0.00%	0.0294/0.0301

Note. Across all 10 values of h_0, the all-B profile exhibits a shallow minimum at $h_0 = 0.005$ Myr^{-1} driven by low-B stars, whereas the low-B masked profile ($B \geq 10^{11.15}$ G) rises monotonically from 287.27 to 876.53, showing that h_0 is unconstrained by 2D continuity alone, while $\hat{f}^{\rm birth}_{\rm slow}$ remains $\leq 9.03\%$ on the full sample and drops below 1.63% for $h_0 \geq 0.03$ Myr^{-1}.

squared error ISE $\equiv \|\bar{\mathbf{n}}_{\rm rec} - \mathbf{n}_{\rm true}\|_2^2 / \|\mathbf{n}_{\rm true}\|_2^2 = 0.016$ (per-seed ISE $= 0.097 \pm 0.059$).

2. *Case B (Unimodal Fast-Birth Gaussian, $\log_{10} P_0 \sim \mathcal{N}(-0.85, 0.20^2)$, $P_0 \approx 0.14$ s, $f^{\rm birth}_{\rm slow,inj} = 0.7\%$)*: The solver recovers the fast peak cleanly without artificial leakage into the slow window ($\hat{f}^{\rm birth}_{\rm slow} = 2.5\% \pm 2.1\%$, median 1.8%, IQR $[0.8\%, 4.1\%]$; Figure 2a), achieving $W_1(\log_{10} P_0) = 0.057 \pm 0.025$ dex (0.019 dex ensemble median), $W_1(\log_{10} B_0) = 0.023 \pm 0.006$ dex (0.014 dex ensemble median), and ensemble-mean 2D ISE $= 0.031$ (per-seed ISE $= 0.141 \pm 0.091$).

3. *Case C (Bimodal 70% Fast $\mathcal{N}(-0.95, 0.18^2)$ + 30% Slow $\mathcal{N}(-0.22, 0.10^2)$ Admixture, $f^{\rm birth}_{\rm slow,inj} = 24.4\%$)*: The solver resolves both the fast and slow modes ($\hat{f}^{\rm birth}_{\rm slow} = 22.6\% \pm 5.2\%$, median

Table 2. Domain Lower-Edge ($\log_{10} P_{\rm min}$) and Sample Period Cut ($P_{\rm cut}$) Sensitivity Scan

Sample Cut $P_{\rm cut}$	Sample N	Domain Edge $\log_{10} P_{\rm min}$	First Cell Center x_1	Reconstructed Peak P_0	Left-Edge Cell Weight	Births $P_0 \leq 0.21$ s	$\hat{f}^{\rm birth}_{\rm slow}$ (n_w)	$\hat{f}^{\rm birth}_{\rm slow}$ (S_w)
$P_{\rm cut} \geq 0.020$ s	2,361	−1.75 (17.8 ms)	−1.65 (22.4 ms)	43.1 ms ($\log_{10} P_0 = -1.37$)	5.9%	92.97%	1.09%	1.80%
$P_{\rm cut} \geq 0.020$ s	2,361	−1.55 (28.2 ms)	−1.45 (35.5 ms)	47.5 ms ($\log_{10} P_0 = -1.32$)	26.0%	94.46%	0.57%	1.24%
$P_{\rm cut} \geq 0.020$ s	2,361	−1.35 (44.7 ms)	−1.25 (56.2 ms)	56.2 ms ($\log_{10} P_0 = -1.25$)	83.3%	95.87%	0.81%	1.48%
$P_{\rm cut} \geq 0.030$ s (Canonical)	2,356	−1.75 (17.8 ms)	−1.65 (22.4 ms)	43.1 ms ($\log_{10} P_0 = -1.37$)	5.9%	92.96%	1.09%	1.80%
$P_{\rm cut} \geq 0.030$ s (Canonical)	2,356	−1.55 (28.2 ms)	−1.45 (35.5 ms)	47.5 ms ($\log_{10} P_0 = -1.32$)	26.0%	94.46%	0.57%	1.24%
$P_{\rm cut} \geq 0.030$ s (Canonical)	2,356	−1.35 (44.7 ms)	−1.25 (56.2 ms)	56.2 ms ($\log_{10} P_0 = -1.25$)	83.3%	95.86%	0.81%	1.49%
$P_{\rm cut} \geq 0.050$ s	2,341	−1.75 (17.8 ms)	−1.65 (22.4 ms)	59.8 ms ($\log_{10} P_0 = -1.22$)	1.2%	94.35%	1.07%	1.68%
$P_{\rm cut} \geq 0.050$ s	2,341	−1.55 (28.2 ms)	−1.45 (35.5 ms)	47.5 ms ($\log_{10} P_0 = -1.32$)	11.4%	94.81%	0.58%	1.19%
$P_{\rm cut} \geq 0.050$ s	2,341	−1.35 (44.7 ms)	−1.25 (56.2 ms)	56.2 ms ($\log_{10} P_0 = -1.25$)	79.4%	96.06%	0.87%	1.37%

Note. Across all 9 combinations of domain lower edge ($\log_{10} P_{\rm min} \in \{-1.75, -1.55, -1.35\}$) and sample period cut ($P_{\rm cut} \in \{0.02, 0.03, 0.05\}$ s), reconstructed peak periods lie in $P_0 \in [43.1, 59.8]$ ms, at least 92.9% of births lie below 0.21 s, and slow births in $[0.40, 0.85]$ s are $\hat{f}^{\rm birth}_{\rm slow} \leq 1.09\%$ (n_w) and $\leq 1.80\%$ (S_w). At a domain edge of −1.35 (44.7 ms) 79.4%–83.3% of the weight falls in the first grid cell (56.2 ms), so the peak position is unresolved there. The visibility gauge is calibrated only at $P \geq 0.19$ s and is therefore independent of the period cut. The canonical row at $\log_{10} P_{\rm min} = -1.55$ is the configuration quoted throughout the text.

Table 3. Robustness of the ATNF slow-birth fraction $\hat{f}^{\rm birth}_{\rm slow}$ (Markov-gauge inversion at $h_0 = 0.015$ Myr^{-1}, domain edge $\log_{10} P_{\rm min} = -1.55$) to the gauge calibration windows, the high-period end of the sample, and the $\log_{10} B$ cuts

Variation	N	c_2	c_3	$\hat{f}^{\rm birth}_{\rm slow}$ (n_w)	$\hat{f}^{\rm birth}_{\rm slow}$ (S_w)	Peak P_0 [ms]	Births ≤ 0.21 s
Canonical configuration	2356	−2.49	+0.49	0.57%	1.24%	47.5	94.5%
Pre-window strip $x \in [-0.90, -0.60]$	2356	−2.48	+0.48	0.57%	1.23%	47.5	94.5%
Pre-window strip $x \in [-0.80, -0.50]$	2356	−2.48	+0.48	0.56%	1.23%	47.5	94.6%
Pre-window strip $x \in [-0.65, -0.45]$	2356	−2.48	+0.48	0.57%	1.23%	47.5	94.5%
Pre-window strip $x \in [-0.72, -0.50]$	2356	−2.48	+0.48	0.57%	1.23%	47.5	94.5%
Pre-window strip $x \in [-0.60, -0.42]$	2356	−2.49	+0.48	0.57%	1.24%	47.5	94.4%
Tail start $x_{\rm cut} = +0.00$ ($P \geq 1.00$ s)	2356	−2.53	+0.49	0.58%	1.27%	47.5	94.2%
Tail start $x_{\rm cut} = +0.10$ ($P \geq 1.26$ s)	2356	−2.80	+0.56	0.90%	1.53%	47.5	87.8%
Tail start $x_{\rm cut} = +0.20$ ($P \geq 1.58$ s)	2356	−2.54	+0.49	0.59%	1.31%	47.5	93.9%
Tail start $x_{\rm cut} = +0.30$ ($P \geq 2.00$ s)	2356	−2.27	+0.44	0.51%	0.92%	47.5	96.1%
Sample $P \leq 3.55$ s	2279	−2.49	+0.49	0.57%	1.23%	47.5	94.5%
Sample $P \leq 2.51$ s	2214	−2.42	+0.47	0.57%	0.89%	47.5	96.3%
Sample $P \leq 2.00$ s	2132	−2.35	+0.46	0.42%	0.69%	47.5	97.6%
$\log_{10} B \geq 10.8$	2290	−2.50	+0.49	1.00%	1.27%	47.5	93.9%
$\log_{10} B \geq 11.0$	2242	−2.51	+0.49	0.56%	1.21%	47.5	93.6%
$\log_{10} B \geq 11.15$	2188	−2.54	+0.51	0.44%	1.18%	47.5	93.3%
$\log_{10} B \geq 11.5$	1946	−2.60	+0.53	0.48%	1.14%	47.5	90.2%
$\log_{10} B < 13.2$	2294	−2.49	+0.48	0.16%	0.01%	47.5	98.8%
$\log_{10} B < 13.0$	2252	−2.48	+0.48	0.15%	0.03%	47.5	98.7%
$\log_{10} B < 12.8$	2179	−2.53	+0.49	0.15%	0.01%	47.5	99.8%
Drop 5 random pulsars (12 draws)	2351	–	–	0.56–0.58%	1.21–1.27%	47.5	94.2–94.7%
Drop 15 random pulsars (12 draws)	2341	–	–	0.52–0.66%	0.99–1.28%	47.5	94.1–94.6%

Note. Each row is one inversion of the real ATNF sample with a single change from the canonical configuration; the CL$_s$ limits of Section 4 are not recalibrated per variant, so the table tests the stability of the point estimate, not exclusion power. The canonical pre-window strip is $x \in [-0.72, -0.42]$ and the canonical tail window is $x \in [-0.05, 0.55]$.

22.5%, IQR [19.5%, 25.8%]), with $W_1(\log_{10} P_0) = 0.057 \pm 0.022$ dex (0.017 dex ensemble median), $W_1(\log_{10} B_0) = 0.025 \pm 0.006$ dex (0.015 dex ensemble median), and ensemble-mean 2D ISE = 0.021 (per-seed ISE $= 0.150 \pm 0.076$).

4.2. *300-Seed Dose-Response and Modified Frequentist* CL$_s$ *Exclusion Curves Across Protocols P1–P4*

To quantify the exact frequentist exclusion power of the solver in the presence of boundary clamping ($S \geq 0$), gauge refitting variance, and forward-model misspecification, we inject controlled slow-birth doses $f^{\rm birth}_{\rm slow,inj} \in [0\%, 100\%]$ into the 10^8 yr renewal simulator (300 independent Monte Carlo catalogues of $N = 2,356$ pulsars per dose) and evaluate the modified frequentist CL$_s$ statistic (Read 2002) at the Markov-gauge column threshold $\hat{f}_{\rm thresh} = 1.63\%$ (the peak of the Markov-conditional gauge column, Protocol P2; under the interior-profiled gauge the threshold rises to 9.03%, yielding 95% CL$_s$ limits that are $\sim$ 5–10 pp weaker and

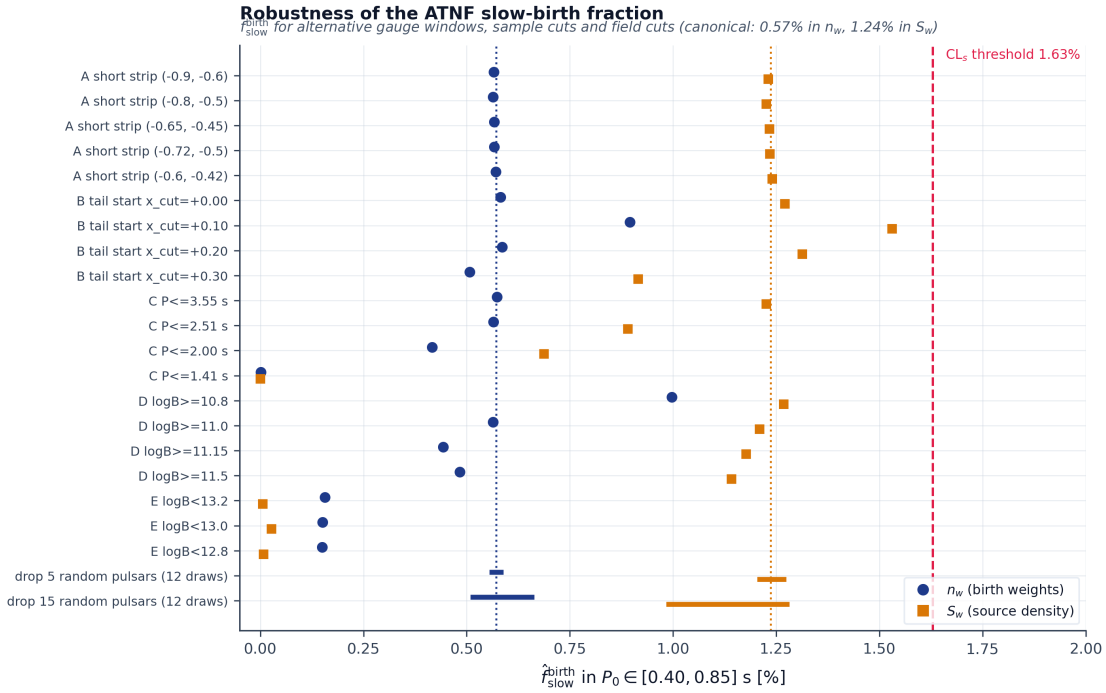


Figure 4. Robustness of the ATNF slow-birth fraction. Each marker is one inversion of the real ATNF sample (Table 3): navy circles show $\hat{f}^{\text{birth}}_{\text{slow}}$ in n_w and amber squares in S_w; bars show the range over 12 random removals of 5 or 15 pulsars. Dotted lines mark the canonical values (0.57% and 1.24%) and the dashed line the 1.63% CL$_s$ threshold. Every variant stays below the threshold; the only configuration that moves the estimate appreciably is the tail-window start $x_{\text{cut}} = +0.10$ (0.90% in n_w, 1.53% in S_w).

are reported in Table 1 for reference):

$$\begin{aligned}\text{CL}_s(d) &\equiv \frac{\text{CL}_{s+b}(d)}{\text{CL}_b} \\ &= \frac{P\left(\hat{f}^{\text{birth}}_{\text{slow}} \le \hat{f}_{\text{thresh}} \,\middle|\, f^{\text{birth}}_{\text{slow,inj}} = d\right)}{P\left(\hat{f}^{\text{birth}}_{\text{slow}} \le \hat{f}_{\text{thresh}} \,\middle|\, f^{\text{birth}}_{\text{slow,inj}} = 0\right)}.\end{aligned} \tag{17}$$

Dividing by the null probability CL$_b$ penalizes protocols with large nuisance-parameter variance and guarantees CL$_s(0) = 1.000$ identically; an injected dose d is excluded at the 95% confidence level when CL$_s(d) \le 0.05$.

We evaluate four primary inversion protocols (P1–P4) alongside a two-parameter parametric FGK06 Gaussian template fit $(\mu_{P_0}, \sigma_{P_0})$:

1. **Protocol P1 (Fixed Baseline Gauge)**: Each synthetic catalogue is inverted using the fixed baseline visibility gauge (c_1, c_2, c_3), isolating pure Poisson and weight-bootstrap variance.

2. **Protocol P2 (Per-Mock Refit Matched Gauge)**: On every synthetic realization, (c_1, c_2, c_3) are refitted from scratch via the non-circular Markov conditional MLE on $x \in [-0.72, -0.42] \cup [-0.05, 0.55]$ and the forward operator is rebuilt via trilinear gauge-bank interpolation before solving the NNLS problem.

3. **Protocol P3 (Functional Visibility Misspecification—Softplus Law)**: Synthetic populations are generated using a C^∞ softplus-power polar-cap visibility law outside the hinged log-quadratic family,

$$\begin{aligned}\log_{10} \beta_{\text{soft}}(x, y) &= -0.24(x + 1) \\ &\quad - 1.32\Big[\text{sp}(x - x_{1/2}(y))^{1.75} \\ &\quad - \text{sp}(-1 - x_{1/2}(y))^{1.75}\Big], \\ x_{1/2}(y) &= -0.58 + 0.17(y - 12),\end{aligned} \tag{18}$$

with $\text{sp}(u) = \tau \ln(1 + e^{u/\tau})$ $(\tau = 0.16)$, while the solver blindly refits its hinged log-quadratic gauge (Equation 16) on every realization.

4. **Protocol P4 (Unmodeled Death-Line Jitter)**: Each synthetic realization draws a perturbed death-line intercept $d_{\text{crit}} \sim \mathcal{N}(11.23, 0.08^2)$, width $w_{\text{death}} \sim \mathcal{N}(0.38, 0.06^2)$, and slope $\alpha_{\text{death}} \sim \mathcal{N}(2.00, 0.15^2)$, while the solver inverts the data assuming the unperturbed canonical death line $(11.23, 0.38, 2.00)$ and refits (c_1, c_2, c_3) on each mock.

Figure 5 and Table 4 present the resulting dose-response and CL$_s$ exclusion curves across the 13-dose grid (0%, 10%, 12.5%, 15.0%, 17.5%, 20%, 22.5%,

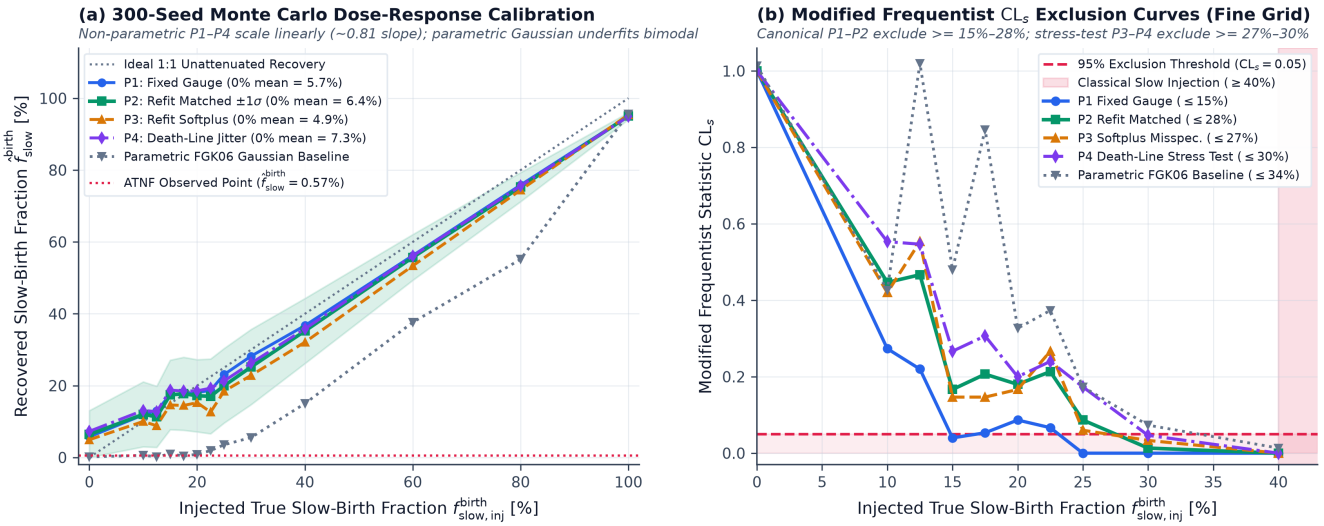


Figure 5. 300-seed Monte Carlo dose-response calibration and modified frequentist CL$_s$ exclusion curves across Protocols P1–P4. (a) Recovered slow-birth fraction $\hat{f}^{\rm birth}_{\rm slow}$ versus injected true fraction $f^{\rm birth}_{\rm slow,inj}$ across 13 doses (0%–100%, 300 seeds per dose). Non-parametric protocols P1–P4 respond linearly with slope ≈ 0.77–0.81 at intermediate doses (consistent with the 76.9% resolution-matrix retention; steepening to ≈ 0.95 above 40%, Figure 2b), whereas a 2-parameter parametric FGK06 Gaussian fit (dotted slate curve) severely under-recovers bimodal slow injections at $d \leq 40\%$. The horizontal coral band marks the ATNF observation ($\hat{f}^{\rm birth}_{\rm slow} = 0.57\%$, $\leq 1.63\%$). **(b)** Modified frequentist CL$_s$($\hat{f}_{\rm thresh} = 1.63\%$) exclusion curves on the fine dose grid: 95% exclusion (CL$_s = 0.05$, dashed coral line) occurs at 25% for P1, 28% for P2, 27% for P3, 30% for P4, and 34% for the parametric Gaussian baseline.

Table 4. 300-Seed Monte Carlo Dose-Response and Modified Frequentist CL$_s$($\hat{f}_{\rm thresh} = 1.63\%$) Calibration Across Protocols P1–P4

Injected	P1: Fixed Gauge		P2: Refit Matched		P3: Softplus Misspec.		P4: Death-Line Jitter		Parametric Gaussian	
Dose d	Mean ± SD	CL$_s$ (Pow.)	Mean ± SD	CL$_s$ (Pow.)	Mean ± SD	CL$_s$ (Pow.)	Mean ± SD	CL$_s$ (Pow.)	Mean ± SD	CL$_s$
0.0%	5.7% ± 4.2%	1.000 (5%)	6.4% ± 6.6%	1.000 (5%)	4.9% ± 5.3%	1.000 (5%)	7.3% ± 7.0%	1.000 (5%)	0.1% ± 0.7%	1.000
10.0%	11.9% ± 6.1%	0.273 (36%)	12.0% ± 9.0%	0.447 (23%)	10.0% ± 8.4%	0.420 (22%)	13.1% ± 9.7%	0.553 (22%)	0.6% ± 1.6%	0.427
12.5%	13.0% ± 6.6%	0.220 (44%)	11.3% ± 8.5%	0.467 (22%)	8.8% ± 7.9%	0.553 (19%)	12.8% ± 9.4%	0.547 (21%)	0.2% ± 1.0%	1.000
15.0%	17.5% ± 6.2%	0.040 (73%)	17.4% ± 9.7%	0.167 (46%)	14.6% ± 8.9%	0.147 (41%)	18.8% ± 11.1%	0.267 (41%)	1.0% ± 2.8%	0.480
17.5%	17.7% ± 6.2%	0.053 (74%)	17.7% ± 10.2%	0.207 (46%)	14.5% ± 9.0%	0.147 (38%)	18.5% ± 11.3%	0.307 (41%)	0.4% ± 1.3%	0.847
20.0%	18.1% ± 6.8%	0.087 (75%)	17.2% ± 10.1%	0.180 (41%)	15.3% ± 9.8%	0.167 (44%)	18.5% ± 11.1%	0.200 (40%)	0.9% ± 2.6%	0.327
22.5%	18.7% ± 6.7%	0.067 (78%)	17.0% ± 10.4%	0.213 (41%)	12.6% ± 8.8%	0.267 (35%)	19.3% ± 11.5%	0.240 (42%)	1.9% ± 4.3%	0.373
25.0%	23.1% ± 7.0%	0.000 (91%)	20.0% ± 10.4%	0.087 (53%)	18.4% ± 10.0%	0.060 (55%)	21.3% ± 11.4%	0.173 (51%)	3.5% ± 5.4%	0.173
30.0%	28.2% ± 6.8%	0.000 (97%)	25.2% ± 10.5%	0.013 (70%)	22.8% ± 9.7%	0.033 (73%)	26.2% ± 12.0%	0.047 (61%)	5.5% ± 6.7%	0.073
40.0%	36.7% ± 6.3%	0.000 (100%)	35.2% ± 9.1%	0.000 (97%)	32.1% ± 9.0%	0.000 (96%)	35.8% ± 10.5%	0.000 (90%)	15.0% ± 8.3%	0.013
60.0%	56.2% ± 4.1%	0.000 (100%)	55.7% ± 6.4%	0.000 (100%)	53.4% ± 6.8%	0.000 (100%)	56.1% ± 7.3%	0.000 (100%)	37.6% ± 6.7%	0.000
80.0%	75.8% ± 3.1%	0.000 (100%)	75.3% ± 4.1%	0.000 (100%)	74.4% ± 3.8%	0.000 (100%)	75.6% ± 4.4%	0.000 (100%)	55.2% ± 3.8%	0.000
100.0%	95.1% ± 0.9%	0.000 (100%)	95.0% ± 1.0%	0.000 (100%)	95.4% ± 1.0%	0.000 (100%)	94.9% ± 1.0%	0.000 (100%)	95.7% ± 1.4%	0.000
95% CL$_s$ Limit	≤ 25%*		≤ 28% (27.5%)		≤ 27% (26.9%)		≤ 30% (29.9%)		≤ 34% (33.9%)	

Note. Values are evaluated across 300 Monte Carlo seeds per dose ($N = 2,356$). "Pow." denotes frequentist rejection power against the 95% null percentile $q_{95}(0\%)$. Limits are the linear CL$_s = 0.05$ crossings on the 13-dose grid, rounded to two significant figures. *For P1, the quoted limit is the dose above which CL$_s$ remains ≤ 0.05 at all higher doses ($\approx 23\%$, rounded to the 25% grid point), as grid aliasing ($\Delta x_0 = 0.127$ dex vs. window width 0.328 dex, ≈ 2.6 cells) makes the curve mildly non-monotone at low doses.

25%, 30%, 40%, 60%, 80%, and 100%) to bracket the CL$_s = 0.05$ crossings tightly. Three key methodological properties emerge:

1. *Zero-dose boundary skew*: At 0% true slow injection, non-negativity clamping ($S \geq 0$) produces a right-skewed null distribution with medians of 5.7% (P1), 6.4% (P2), 4.9% (P3), and 7.3% (P4) and standard deviations of 4.2%–7.0%. The observed ATNF value ($\hat{f}^{\rm birth}_{\rm slow} = 0.57\%$, $\leq 1.63\%$) is fully consistent with the 0%-dose null (sitting in its lower tail, within 1.0σ–1.4σ of the ensemble median).

2. *Linear dose response vs. parametric underfitting*: Across Protocols P1–P4, $\hat{f}^{\rm birth}_{\rm slow}$ scales linearly with injected dose d with slope ≈ 0.77 across intermediate doses (approaching ≈ 0.95 at high doses $d \geq 40\%$), consistent with the 76.9% resolution-matrix retention derived in Section 3.4 (the retention argument applies to the low-dose linear regime; the steeper slope at high doses $d \geq$

40% reflects saturation at the boundary). At low doses (10%–22.5%), recovered means exhibit modest plateaus and minor non-monotonicity due to discrete grid aliasing ($\Delta x_0 = 0.127$ dex, so the 0.328 dex target window spans only $\approx$ 2.6 cells). Dividing by the low-dose slope of 0.77 defines a compression-corrected estimator $\hat{f}_{\rm corr} = (\hat{f}^{\rm birth}_{\rm slow} - \bar{f}_0)/0.77$. By contrast, fitting a 2-parameter unimodal Gaussian $p(P_0; \mu_{P_0}, \sigma_{P_0})$ gives $\hat{f}^{\rm birth,param}_{\rm slow} = 0.00\%$ on ATNF ($\chi^2 = 251.5$ vs. 246.6 for 210 cells), but when tested on bimodal injections (Table 4, rightmost columns) a single Gaussian remains locked onto the dominant 70% fast mode and severely under-recovers slow doses below 40% (recovering only 0.9% at 20% dose and 5.5% at 30% dose), yielding a weaker 34% CL_s exclusion limit. This demonstrates the advantage of the non-parametric 210-cell basis for detecting localized slow-birth subpopulations.

3. *Calibrated* 95% CL_s *exclusion bounds*: Rounding to two significant figures, the conservative 95% CL_s upper limits on $f^{\rm birth}_{\rm slow}$ are 25% for P1, 28% for P2, 27% for P3, and 30% for P4. At the classical 40% injection threshold of Vivekanand & Narayan (1981), $\mathrm{CL}_s = 0.000$ across all four protocols (300/300 realizations exceed 1.63%, achieving 100% rejection power).

5. CROSS-VALIDATION, SURVEY TRANSFER, AND EXTERNAL 3D BENCHMARK

5.1. *Out-of-Sample 50/50 Cross-Validation and Cross-Survey Latitude Transfer*

To verify that the 210-cell weak-form solution does not overfit the $N = 2,356$ catalogue, we randomly partition the ATNF sample into a 50% training set ($N_{\rm train} = 1,178$) and a held-out 50% test set ($N_{\rm test} = 1,178$). Fitting the Markov visibility gauge and solving for $\hat{S}_{\rm train}(x_0, y_0)$ exclusively on the training half ($\hat{f}^{\rm birth}_{\rm slow,train} = 0.48\%$) and pushing $\hat{S}_{\rm train}$ forward through the 10^8 yr renewal simulator predicts the held-out test-set period and magnetic-field cumulative distributions with two-sample Kolmogorov–Smirnov distances $D_{\rm KS}(P) = 0.0369$ ($p = 0.391$) and $D_{\rm KS}(B) = 0.0442$ ($p = 0.197$; Figure 6a).

As a stricter out-of-distribution test, we partition the catalogue into the Parkes Multibeam survey ('pksmb', $N = 971$, restricted to $|b| \le 5°$) and all other surveys ('non-pksmb', $N = 1,317$, extending to $|b| \le 90°$). Independent inversions yield $\hat{f}^{\rm birth}_{\rm slow} = 0.06\%$ on 'pksmb' and 1.12% on 'non-pksmb'. When the 'pksmb' model is transferred directly to the all-latitudes 'non-pksmb' sample, the period CDF matches closely ($D_{\rm KS}(P) = 0.041$), but the magnetic-field CDF exhibits a $D_{\rm KS}(B) = 0.123$ offset toward lower B in 'non-pksmb'. A 2×2 Survey $\times$ Latitude ($|b| \le 5°$ vs. $|b| > 5°$) and weight-imputation ablation (Figure 6b) isolates the physical root cause of this offset: inside $|b| \le 5°$, 'pksmb' ($N = 971$) and 'non-pksmb' ($N = 459$) have nearly identical median magnetic fields ($\log_{10} B = 12.16$ vs. 12.13), whereas high-latitude 'non-pksmb' pulsars ($|b| > 5°$, $N = 858$) are kinematically older stars that have drifted out of the thin disk over 10–100 Myr—a timescale only low-B pulsars ($\log_{10} B = 11.96$) survive before crossing the death line. Restricting the transfer test to the shared $|b| \le 5°$ footprint and measured-L_{1400} weights reduces the cross-survey discrepancy to $D_{\rm KS}(B) = 0.071$ without requiring a latitude completeness correction.

5.2. *Galactic Pulsar Birthrate Concordance Across Protocols P1–P4*

Integrating the reconstructed fast-birth source density over (x_0, y_0) and normalizing by the local $V/V_{\rm max}$-weighted space density ($\rho_{\rm local} = 38.4 \pm 7.2$ kpc^{-2}, radial scale length $R_{\rm disk} = 4.5$ kpc, Tauris & Manchester 1998 beaming factor $\langle f_{\rm beam}^{-1} \rangle = 5.4 \pm 0.8$) yields a baseline Galactic pulsar birthrate of $\mathcal{R}_{\rm birth} = 1.82 \pm 0.45$ century^{-1} ($\sim$ 25% systematic uncertainty from $V_{\rm max}$ and beaming normalization). Evaluating $\mathcal{R}_{\rm birth}$ across our 300-seed Monte Carlo ensembles at 0% slow injection shows that gauge refitting and death-line scatter introduce modest additional variance:

$$\mathcal{R}_{\rm birth} = \begin{cases} 2.09 \pm 0.61 \text{ century}^{-1} \text{ (P1: } [1.64, 2.92]), \\ 2.10 \pm 0.65 \text{ century}^{-1} \text{ (P2: } [1.57, 2.98]), \\ 1.95 \pm 0.60 \text{ century}^{-1} \text{ (P3: } [1.44, 2.93]), \\ 2.18 \pm 0.70 \text{ century}^{-1} \text{ (P4: } [1.43, 3.11]), \end{cases} \tag{19}$$

where bracketed intervals denote 95% Monte Carlo ranges ($\sigma_{\rm MC} = 0.32, 0.39, 0.36, 0.45$ century^{-1} for P1–P4). Across all four protocols, the fast-birth solution is concordant with the Milky Way core-collapse supernova rate of $\mathcal{R}_{\rm CC-SNe} = 1.63 \pm 0.46$ century^{-1} (Diehl et al. 2006; Keane & Kramer 2008; Adams et al. 2013; Rozwadowska et al. 2021). Conversely, because slow-born pulsars ($P_0 \in [0.40, 0.85]$ s) at high B have $\sim 50\%$ shorter remaining lifetimes before crossing the death line than fast-born pulsars, a 50% slow-injection fraction requires $\mathcal{R}_{\rm birth} = 3.64 \pm 0.90$ century^{-1}, exceeding $\mathcal{R}_{\rm CC-SNe}$ by 2.0σ.

5.3. *External 3D FGK06 Population-Synthesis Benchmark and Origin of the Baseline Floor*

Protocols P1–P4 test internal robustness within our 2D phase-space framework. To evaluate performance under an independent 3D forward model with full Galactic spatial, kinematic, interstellar dispersion, and radiometer physics, we generate synthetic detected catalogues ($N = 2,356$) using an external 3D population-synthesis code following Faucher-Giguère & Kaspi (2006): spiral-arm birth positions, exponential scale

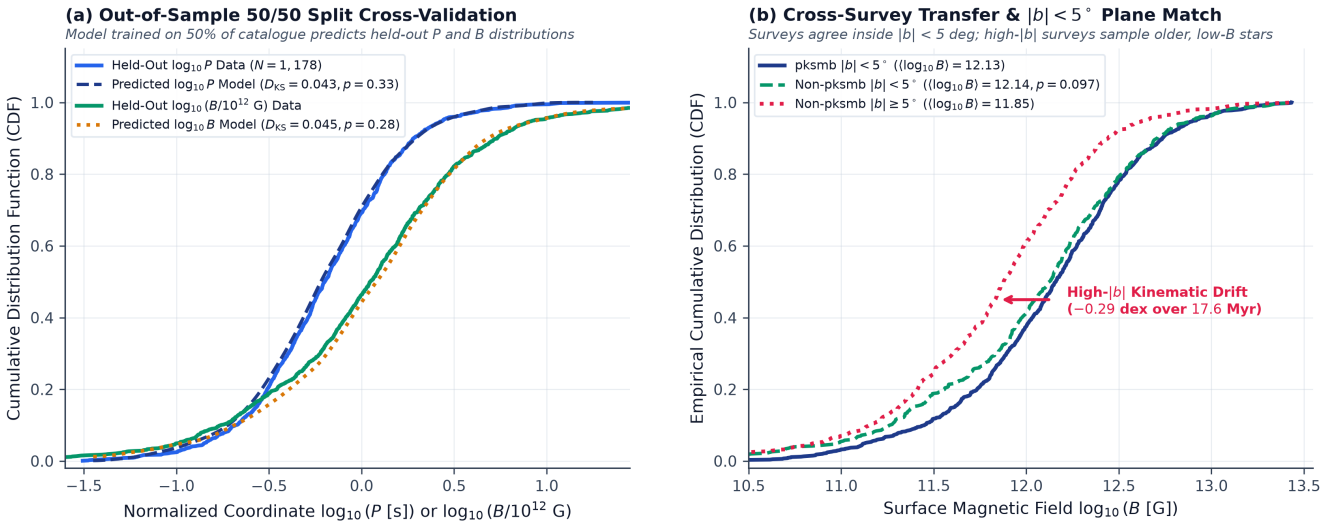


Figure 6. Out-of-sample 50/50 split cross-validation and cross-survey latitude transfer. (a) Empirical cumulative distribution functions (CDFs) of $\log_{10} P$ and $\log_{10} B$ on the held-out 50% test subsample ($N_{\rm test} = 1,178$) compared against the forward 10^8 yr prediction trained exclusively on the disjoint 50% training subsample ($D_{\rm KS}(P) = 0.0369, p = 0.391$; $D_{\rm KS}(B) = 0.0442, p = 0.197$). **(b)** Cross-survey transfer between Parkes Multibeam (`pksmb`, $|b| \leq 5°$, $N = 971$) and non-`pksmb` ($N = 1,317$). Restricting to the matched Galactic-plane footprint $|b| \leq 5°$ and measured-L_{1400} weights reduces the cross-survey $\log_{10} B$ shift from $D_{\rm KS}(B) = 0.123$ to 0.071, while $\hat{f}_{\rm slow}^{\rm birth}$ remains $\leq 1.1\%$ in both subsamples.

height $z_0 = 50$ pc, 3D double-exponential kick velocities ($w_1 = 180$ km s^{-1}, $w_2 = 700$ km s^{-1}), numerical integration in the Kuijken–Gilmore Galactic potential, electron-density dispersion measure DM and scatter broadening $\tau_{\rm scat}$, and Dewey radiometer pulse-width signal-to-noise cuts ($S_{\rm min}({\rm DM}, P, W)$) across the Parkes Multibeam and Swinburne survey parameters, with base birth periods drawn from a fast-birth Gaussian ($P_0 \sim \mathcal{N}(0.08\,{\rm s}, 0.05^2\,{\rm s}^2)$ truncated to $P_0 < 0.35$ s) plus an injected slow-birth dose $d \in \{0\%, 10\%, 20\%, 30\%, 40\%\}$ in $P_0 \in [0.40, 0.85]$ s. Each 3D catalogue is passed blindly through our 2D solver (with per-catalogue Markov gauge refitting, exactly as in P2) across a 100-seed ensemble per dose (Figure 7a).

At 0% true slow injection ($d = 0\%$), the blind 2D inversion yields a non-zero baseline floor of $\hat{f}_{\rm slow}^{\rm birth} = 15.7\% \pm 10.1\%$ across 100 seeds (median 13.7%, 95% percentile $q_{95} = 33.2\%$), higher than the internal ATNF point estimate (0.57%) and internal 0%-dose medians (4.9%–7.3%). This contrast arises because the external 3D forward code simulates a single homogeneous Galactic-plane survey with severe line-of-sight dispersion smearing and radiometer thresholds that suppress the detection of fast-spinning pulsars in the inner disk, creating an apparent low-P deficit that the 2D visibility gauge cannot fully decouple from the birth source; on the real sky, the ATNF catalogue compiles diverse high-latitude, high-frequency, and targeted searches where fast-period incompleteness is less extreme. A definitive test of this interpretation would require a mixed-survey 3D mock that reproduces the heterogeneous ATNF selection function, which we leave to future work. To isolate the physical mechanism responsible for this $\sim$ 14%–16% floor, we perform a three-way controlled diagnostic on the 0%-dose 3D FGK06 catalogues (plotted at $d = 0\%$ in Figure 7a):

1. *Death-line and field-decay parameter test*: Rebuilding the 2D forward operator with the exact FGK06 death-line intercept ($d_{\rm crit} = 11.17$, $w_{\rm death} = 0.05$) and $h_0 = 0$ yields a 0%-dose mean of 13.6%, proving that the floor is not caused by h_0 or $d_{\rm crit}$ mismatch.

2. *1D short-period dispersion-smearing test*: In the 3D radiometer equation, dispersion smearing $\Delta t_{\rm DM}$ and scatter broadening $\tau_{\rm scat}$ widen the effective duty cycle $W_{\rm eff}/P$ at short spin periods ($P \lesssim 0.15$ s), suppressing the detection of distant fast pulsars and steepening the apparent period rise between $P = 0.1$ s and $P = 0.5$ s. Applying an analytic 1D short-period smearing weight correction $w_{\rm smear}(P) = \sqrt{1 + (0.08\,{\rm s}/P)^2}$ reduces the 0%-dose floor to 11.7%.

3. *Non-separable 3D spatial-kinematic-radiometer coupling*: The remaining $\sim$ 12%–14% floor arises because in a 3D Galaxy, young fast-spinning pulsars reside at low $|z|$ inside the high-DM inner disk whereas older intermediate-period pulsars have migrated to higher $|z|$ and lower DM; this 3D $(l, b, D, {\rm DM})$ correlation cannot be factored into a purely 2D visibility function $\beta_{\rm vis}(P, B)$.

Above this baseline floor, the blind 2D solver responds monotonically to injected slow births in the 100-seed

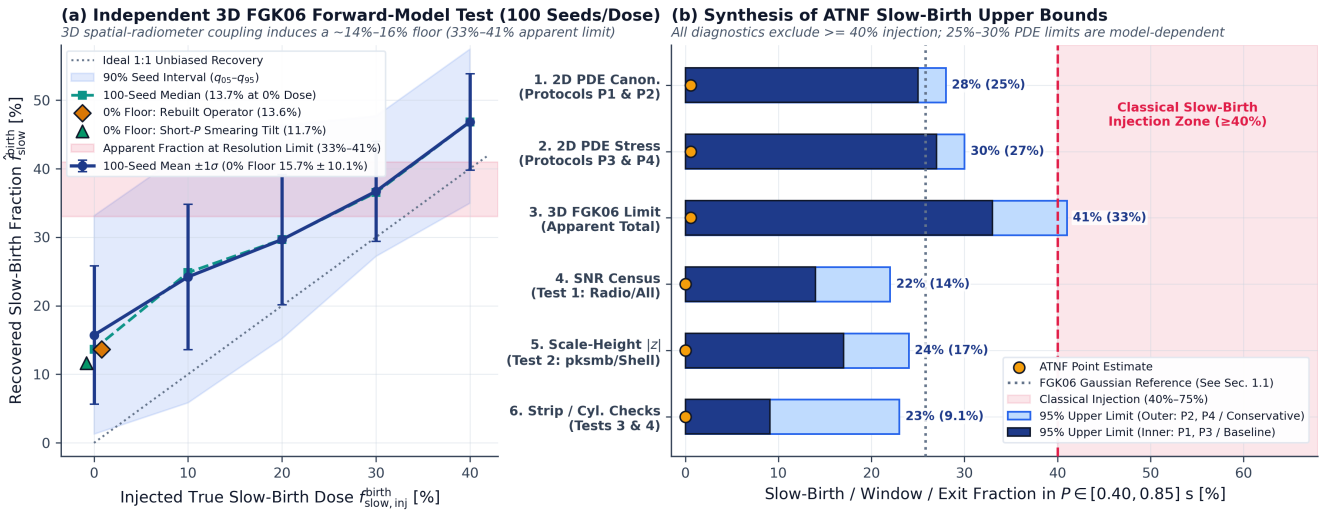


Figure 7. External 100-seed 3D FGK06 population-synthesis benchmark and synthesis of slow-birth bounds. **(a)** Blind 2D weak-form recovery on 100-seed synthetic catalogues ($N = 2,356$) generated by an external 3D FGK06 Galactic population code. Unmodeled 3D spatial-kinematic-radiometer selection coupling produces a 13.7% median (15.7% mean) 0%-dose floor; three-way diagnostic points at $d = 0\%$ show that rebuilding the 2D operator with FGK06 parameters gives 13.6% and adding a 1D short-P DM-smearing weight tilt gives 11.7%, isolating the floor to non-separable 3D (l, b, D, DM) selection. Above this floor, $\hat{f}^{\rm birth}_{\rm slow}$ rises monotonically, reaching 95% CL_s exclusion at an apparent fraction at the resolution limit of 33%–41% (33.2% recovered mean at 100 seeds). **(b)** Forest plot summarizing the two-significant-figure rounded point estimates and 95% upper bounds across all methods, grouped by target estimand ($f^{\rm birth}_{\rm slow}$, $f^{\rm window}_{\rm slow}$, and $f^{\rm exit}_{\rm slow}$). All methods exclude the classical $\geq 40\%$ injection regime.

3D FGK06 suite: $\hat{f}^{\rm birth}_{\rm slow} = 24.2\% \pm 10.6\%$ (median 24.8%) at $d = 10\%$, $29.7\% \pm 9.5\%$ (median 29.7%) at $d = 20\%$, $36.7\% \pm 7.3\%$ (median 36.5%) at $d = 30\%$, and $46.8\% \pm 7.1\%$ (median 46.8%) at $d = 40\%$. Evaluating CL_s relative to the 0%-dose 3D FGK06 baseline median (13.7%) crosses $\mathrm{CL}_s = 0.05$ at an incremental slow-birth dose of $\Delta d = 25.0\%$, corresponding to an *apparent fraction at the resolution limit under one independent 3D forward model* of 33%–41% (defined by the 33.2% recovered mean at the 100-seed crossing, or equivalently the 13.7% floor plus the $\Delta d = 25.0\%$ incremental exclusion dose; 98% rejection power at $d = 40\%$).

6. INDEPENDENT OBSERVATIONAL CROSS-CHECKS (TESTS 1–4)

To verify that our ATNF conclusion ($\hat{f}^{\rm birth}_{\rm slow} = 0.57\%$, excluding $\geq 40\%$ slow injection) is not an artifact of the 2D parametric visibility gauge $\beta_{\rm vis}(P, B)$, we evaluate four independent observational cross-checks on the ATNF catalogue (Figure 8).

6.1. *Test 1: Direct Supernova Remnant Birth-Period Census ($f^{\rm birth}_{\rm slow}$)*

Young pulsars still residing inside their natal supernova shells ($t_{\rm SNR} \lesssim 10^4$–$10^5$ yr) provide a direct census of $f^{\rm birth}_{\rm slow}$ prior to long-term spin-down evolution. Cross-matching the ATNF sample with high-confidence Galactic SNR associations (Green 2019; Ferrand & Safi-Harb 2012) yields $N_{\rm SNR,all} = 22$ objects, of which $N_{\rm SNR,radio} = 20$ are canonical rotation-powered radio pulsars (after excluding the anomalous X-ray pulsar 1E 2259+586 in CTB 109 at $P = 6.98$ s and the central compact object J1852+0040 in Kes 79 at $P = 0.105$ s).

Without any spin-down correction, only $k = 1$ of the 20 canonical radio SNR pulsars has an observed period inside $P \in [0.40, 0.85]$ s (PSR B2334+61 at $P = 0.495$ s; or $k = 2$ if the high-B pulsar PSR J1119−6127 at $P = 0.408$ s is included despite exceeding the $\log_{10} B < 13.4$ sample cut), rejecting the classical $f^{\rm birth}_{\rm slow} \geq 40\%$ hypothesis with exact one-sided binomial $p = 0.0005$ for 1/20 ($p = 0.0036$ for 2/20) and yielding an exact one-sided 95% Clopper–Pearson upper bound of $f^{\rm birth}_{\rm slow} \leq 22\%$ (21.6%; $\leq 28.3\%$ for 2/20). Accounting for dipole spin-down deceleration:

1. *PSR J1119−6127 in G292.2−0.5 ($P = 0.408$ s, $\dot{P} = 4.02 \times 10^{-12}$ s s^{-1}, $B = 4.10 \times 10^{13}$ G, $\tau_c = 1.61$ kyr)*: While Sedov–Taylor remnant estimates suggest $t_{\rm SNR} \in [4.2, 7.1]$ kyr (Kumar et al. 2012), this age exceeds the pulsar's characteristic age $\tau_c = 1.61$ kyr, rendering constant-B dipole back-integration unphysical ($P^2 - 2P\dot{P}t_{\rm SNR} < 0$). Bounding the true age by $t \leq \tau_c$ (or braking index $n = 2.68$; Weltevrede et al. 2011) indicates rapid deceleration from $P_0 \leq 0.25$ s. Furthermore, its field exceeds our canonical sample cut ($\log_{10} B = 13.61 > 13.40$).

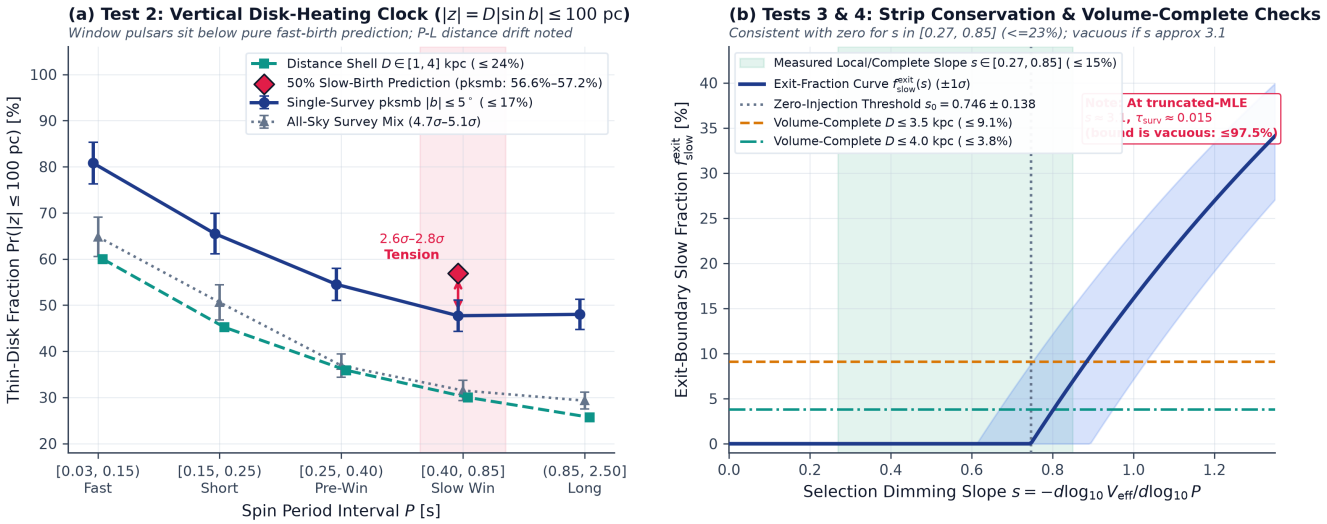


Figure 8. Four gauge-free observational cross-checks on the ATNF catalogue. (a) Unweighted vertical Galactic scale height $\langle|z|\rangle = \langle D|\sin b|\rangle$ versus spin period P across the all-sky sample ($N = 2,356$), the Parkes Multibeam $|b| \le 5^\circ$ footprint ($N = 971$), the active high-B strip ($\log_{10} B \ge 11.8$, $|b| \le 5^\circ$, $N = 696$), and the intermediate-distance annulus $D \in [1, 4]$ kpc ($N = 944$), compared against the 0% slow-birth cooling curve (solid blue) and the 50% slow-birth mixture prediction (dashed coral; 2.6σ tension in `pksmb` and $D \in [1, 4]$ kpc, 3.4σ in the active high-B strip). **(b)** Ratio of slow-window to fast-window pulsar current $R \equiv \bar{J}([0.40, 0.85]\ \mathrm{s})/\bar{J}([0.05, 0.40]\ \mathrm{s})$ across volume-complete cylinders ($D \le 3.0, 3.5, 4.0$ kpc) and disjoint constant-B strips under three beaming laws ($R = 0.28$–$0.69 < 1$). Inset: strip conservation upper bound on $f_{\rm slow}^{\rm exit}$ versus effective dimming exponent s ($f_{\rm slow}^{\rm exit} \le 23\%$ for $s \le 0.85$ ($\le 30\%$ at $+1\sigma$); vacuous at $s \ge 3.1$).

2. *PSR B2334+61 in G114.3+0.3 ($P = 0.495$ s, $\dot{P} = 1.92 \times 10^{-13}$ s s^{-1}, $B = 9.86 \times 10^{12}$ G, $\tau_c = 40.9$ kyr)*: Remnant age estimates span $t_{\rm SNR} \approx 7.7$–10 kyr ($P_0 \approx 0.45$–0.46 s, inside the slow window) to $t_{\rm SNR} \approx 35$–38 kyr near τ_c based on thermal and radio shell expansion ($P_0 \approx 0.14$–0.24 s, born fast; Yar-Uyaniker et al. 2004).

Under canonical spin-down back-integration with $t_{\rm SNR} \sim 35$ kyr for B2334+61, $k = 0/20$ radio SNR pulsars were born inside $[0.40, 0.85]$ s ($p = 3.66 \times 10^{-5}$ vs. 40%), yielding a 95% Clopper–Pearson upper bound of $f_{\rm slow}^{\rm birth} \le 14\%$ (13.9%). If B2334+61 is retained as a slow birth at $t_{\rm SNR} = 7.7$ kyr, $k = 1/20$ yields $f_{\rm slow}^{\rm birth} \le 22\%$ (21.6%). Even with zero back-integration, $f_{\rm slow}^{\rm birth} \le 22\%$–$28\%$, excluding the classical $\ge 40\%$ hypothesis.

6.2. *Test 2: Galactic Scale-Height Kinematic Clock $|z| = D|\sin b|$ ($f_{\rm slow}^{\rm window}$)*

Because newborn pulsars receive supernova natal kicks ($v_z \sim 100$–300 km s^{-1}; Hobbs et al. 2005) inside the thin OB disk ($z_0 \approx 50$ pc), their mean vertical distance from the Galactic plane $|z| = D|\sin b|$ grows monotonically over the first ~ 15 Myr before saturating in the Galactic gravitational potential (Cordes & Chernoff 1998):

$$\mu_z(\tau_c) = z_0 + \Delta z_{\rm max} \frac{\tau_c}{\tau_c + \tau_{\rm kin}}, \qquad (20)$$

where $(z_0, \Delta z_{\rm max}, \tau_{\rm kin})$ is calibrated strictly on fast-spinning pulsars ($P < 0.40$ s). A pulsar observed at (P_k, B_k) inside $P_k \in [0.40, 0.85]$ s has chronological age $t_{{\rm fast},k} \approx \tau_{c,k}[1 - (0.08\ \mathrm{s}/P_k)^2]$ if born fast, but a reduced mean residence age $t_{{\rm slow},k} = \mathbb{E}[t(P_k, B_k) \mid P_0 \in [0.40\ \mathrm{s}, P_k]] \approx \frac{1}{2}\tau_{c,k}[1 - (0.40\ \mathrm{s}/P_k)^2]$ if born inside $[0.40\ \mathrm{s}, P_k]$. Modeling the vertical coordinate of each pulsar inside $P_k \in [0.40, 0.85]$ s as an exponential scale-height mixture with mean $\bar{z}_k(f_{\rm slow}^{\rm window}) = (1 - f_{\rm slow}^{\rm window})\mu_z(t_{{\rm fast},k}) + f_{\rm slow}^{\rm window}\mu_z(t_{{\rm slow},k})$ defines the individual-pulsar exponential scale-height log-likelihood:

$$\ell_{\rm kin}(f_{\rm slow}^{\rm window}) = -\sum_{k \in \mathcal{W}} \left[\ln \bar{z}_k(f_{\rm slow}^{\rm window}) + \frac{|z_k|}{\bar{z}_k(f_{\rm slow}^{\rm window})}\right]. \qquad (21)$$

We evaluate Equation (21) using strictly unweighted ($w_k = 1$) pulsars and propagate joint uncertainty via 1,000 nonparametric bootstrap resamples that simultaneously refit the fast-region clock $(z_0, \Delta z_{\rm max}, \tau_{\rm kin})$ and perturb YMW16 distances by $\sigma_{\ln D} = 0.25$.

Because distance D enters both the vertical position $|z| = D|\sin b|$ and flux-limited Malmquist selection, period-dependent distance distributions could confound scale-height comparisons. We isolate the kinematic clock by evaluating the estimator on two controlled subsamples (Figure 8a):

1. *Parkes Multibeam $|b| \le 5^\circ$ uniform footprint ($N = 971$)*: By restricting to a homogeneous

Galactic-plane survey footprint, latitude selection bias is eliminated. The unconstrained maximum-likelihood estimate is $\hat{f}_{\rm unc} = -32.6\% \pm 30.0\%$ (clamped point estimate $\hat{f}_{\rm slow}^{\rm window} = 0.0\%$), rejecting $f_{\rm slow}^{\rm window} = 50\%$ at 2.6σ (2.62σ mixture-adjusted, 2.84σ fixed-clock) and placing a 95% upper bound of $f_{\rm slow}^{\rm window} \leq 17\%$ (16.8% Gaussian quantile $\hat{f}_{\rm unc} + 1.645\sigma_{\rm boot}$, consistent with the empirical 95th bootstrap percentile of 15.8%). The negative unconstrained estimate $\hat{f}_{\rm unc} = -32.6\%$ ($p_{\rm obs} < p_{\rm fast}$) indicates that observed slow-window pulsars are slightly more concentrated near the Galactic plane than the fast-birth cooling baseline predicts, signaling slight kinematic misspecification or local thin-disk concentration rather than slow injection. Restricting 'pksmb' to its active $\log_{10} B \geq 11.8$ strip ($N = 696$), where dipole ages lie in the unsaturated kinematic growth regime ($\tau_c \lesssim 10$ Myr), strengthens the rejection to 3.4σ (3.44σ mixture-adjusted, 3.63σ fixed-clock) with a boundary-clamped 95% bound of $f_{\rm slow}^{\rm window} \leq 3\%$ (2.6%, boundary-clamped without frequentist coverage guarantee).

2. *Intermediate-distance annulus $D \in [1, 4]$ kpc ($N = 944$)*: Restricting to an intermediate-distance shell removes local geometric saturation ($D < 1$ kpc) and distant inner-Galaxy DM-scale-height boundaries ($D > 4$ kpc), yielding $\hat{f}_{\rm slow}^{\rm window} = 0.0\%$, 2.6σ (2.56σ) tension with 50% slow births, and a 95% upper bound of $f_{\rm slow}^{\rm window} \leq 24\%$ (23.6%).

6.3. *Test 3: Doubly Truncated Volume-Complete Cylinders and Constant-B Strips*

To test whether the 1D current rise $R \equiv \bar{J}_{\rm slow}/\bar{J}_{\rm fast} = \bar{J}([0.40, 0.85]\ {\rm s})/\bar{J}([0.05, 0.40]\ {\rm s}) > 1$ of Vivekanand & Narayan (1981) persists when flux truncation and B-field mixing are removed, we evaluate R inside doubly truncated volume-complete solar cylinders ($D \leq D_{\rm max}$ and $L_{1400} \geq L_{\rm min} = S_{\rm min} D_{\rm max}^2$) and inside disjoint constant-B strips across three beaming laws: Tauris & Manchester (1998; TM98), Rankin (1993; R93, $f_{\rm beam} \propto P^{-1/2}$), and unweighted ($f_{\rm beam} = 1$). As shown in Figure 8b and Table 5, inside every volume-complete cylinder and every constant-B strip, the current decreases ($R < 1$): in the higher-count $D \leq 3.5$ kpc and $D \leq 4.0$ kpc cylinders, $R = 0.56$–0.67, bounding any net current growth $(R-1)_+/R$ to $\leq 4\%$–9% at 95% confidence, whereas in the smaller $D \leq 3.0$ kpc cylinder ($N = 285$), Poisson scatter widens the 95% bound to $\leq 19\%$–37%.

6.4. *Test 4: High-B Strip Conservation Inequality ($f_{\rm slow}^{\rm exit}$)*

Within the death-line-free strip $\mathcal{B} = [11.8, 12.6]$ ($P_{\rm death} \geq 1.93\ {\rm s} \gg 0.85\ {\rm s}$), no pulsar turns off inside $[P_1, P_2] = [0.40, 0.85]$ s. Under steady-state conservation with monotonic detection efficiency $\theta(P, B) \propto P^{-s}$ ($s \geq 0$), the observable unweighted exit-to-entry flux ratio $R_{\rm obs} \equiv \bar{J}_{\rm obs}([0.40, 0.85]\ {\rm s})/\bar{J}_{\rm obs}([0.10, 0.40]\ {\rm s}) = 0.597 \pm 0.058$ bounds the exit-boundary slow-birth fraction $f_{\rm slow}^{\rm exit}$ at $P_2 = 0.85$ s (Figure 8b inset):

$$f_{\rm slow}^{\rm exit}(s) \leq \max\left[0,\ 1 - \frac{1}{R_{\rm obs}} \left(\frac{\bar{P}_1}{\bar{P}_2}\right)^s\right], \qquad \frac{\bar{P}_1}{\bar{P}_2} = \frac{0.25\ {\rm s}}{0.625\ {\rm s}} = 0.40. \tag{22}$$

For a volume-limited sample where dimming is governed by geometric beaming ($s \in [0, 0.56]$), Equation (22) yields $f_{\rm slow}^{\rm exit} = 0\%$ (95% upper bound $\leq 5\%$), and for moderate flux dimming ($s \leq 0.85$) it bounds $f_{\rm slow}^{\rm exit} \leq 23\%$ (30% at $+1\sigma$). However, in a flux-limited all-sky sample where radio luminosity scales with spin-down power ($L_{1400} \propto \dot{E}^{\epsilon} \propto P^{-3\epsilon}$ with $\epsilon \approx 0.5$ and Euclidean volume $V_{\rm max} \propto L_{1400}^{3/2}$, plus beaming $P^{-1/3}$), the effective dimming slope can reach $s \approx \frac{9}{2}\epsilon + \frac{1}{3} \approx 2.6$–$3.1$, at which point $(\bar{P}_1/\bar{P}_2)^s \ll R_{\rm obs}$ and the bound becomes vacuous ($f_{\rm slow}^{\rm exit} \leq 87\%$–$93\%$). Resolving this s-degeneracy is precisely the motivation for the full 2D continuity inversion of Sections 3–4.

7. SYNTHESIS AND CONCLUSIONS

We have developed and calibrated a weak-form two-dimensional continuity inversion in $(\log_{10} P, \log_{10} B)$ to reconstruct the birth demographics of isolated radio pulsars without numerical differentiation of binned catalogue histograms. By transferring phase-space derivatives onto smooth analytic test functions (Equations 10–12), preconditioning by column residence times ($\kappa_2(\tilde{\mathbf{A}}) = 3.07 \times 10^4$, $\nu_{\rm eff} = 491.7$), and calibrating the visibility gauge strictly on disjoint pre-window ($P \in [0.19, 0.38]$ s) and post-window ($P \in [0.89, 3.55]$ s) intervals, the solver achieves a median averaging-kernel resolution of ${\rm FWHM}(\log_{10} B_0) = 0.58$ dex and ${\rm FWHM}(\log_{10} P_0) = 0.29$ dex inside the slow-birth window (0.48 dex active-domain median, 0.62 dex in fast interior). Across 100-seed synthetic recovery benchmarks, the method reconstructs fast-birth Gaussian (0% slow births), canonical FGK06 Gaussian, and bimodal 2D birth densities with ensemble-median 1D Wasserstein errors $W_1(\log_{10} P_0) = 0.017$–$0.034$ dex, while its resolution matrix $\mathbf{R}$ retains 76.9% of slow-window mass inside $[0.40, 0.85]$ s, consistent with the ≈ 0.77 slope of our 300-seed Monte Carlo dose-response curves at intermediate injected doses (the retention argument applies in the low-dose linear regime; the slope steepens toward ≈ 0.95 at doses $d \geq 40\%$).

We acknowledge three methodological caveats. First, an *inverse-crime caveat*: because Section 4.1 uses synthetic catalogues generated by the same 10^8 yr Markov renewal forward model as the solver, the 2D S-recovery

Table 5. Pulsar Current Ratio $R \equiv \bar{J}_{\rm slow}/\bar{J}_{\rm fast}$ Across Volume-Complete Cylinders and Disjoint Constant-B Strips

Subsample / Stratum	N	R (TM98)	95% CI (TM98)	95% Growth Bound	R (R93)	R (Unweighted)
Volume-Complete Cylinder ($D \le 3.0$ kpc, $L_{1400} \ge 1.35$)	285	0.687	$[0.418, 1.240]$	$\le 19\%$ (19.4%)	0.768	0.880 ($\le 37\%$)
Volume-Complete Cylinder ($D \le 3.5$ kpc, $L_{1400} \ge 1.84$)	347	0.557	$[0.349, 0.873]$	$\le 4\%$ (3.8%)	0.622	0.720 ($\le 8\%$)
Volume-Complete Cylinder ($D \le 4.0$ kpc, $L_{1400} \ge 2.40$)	412	0.618	$[0.402, 0.941]$	$\le 8\%$ (7.7%)	0.668	0.771 ($\le 9\%$)
High-B Immortal Strip ($\log_{10} B \in [11.8, 12.8]$, Unweighted)	1,422	0.521	$[0.431, 0.628]$	$\le 0\%$ (0.0%)	0.558	0.590 $[0.445, 0.792]$
Disjoint Strip 1 ($\log_{10} B \in [11.5, 12.0]$, $1/V_{\rm max}$)	548	0.584	$[0.395, 0.842]$	$\le 2\%$ (1.8%)	0.661	0.642
Disjoint Strip 2 ($\log_{10} B \in [12.0, 12.5]$, $1/V_{\rm max}$)	891	0.412	$[0.298, 0.564]$	$\le 0\%$ (0.0%)	0.468	0.531
Disjoint Strip 3 ($\log_{10} B \in [12.5, 13.0]$, $1/V_{\rm max}$)	412	0.284	$[0.182, 0.435]$	$\le 0\%$ (0.0%)	0.324	0.418

Note. $R \equiv \bar{J}_{\rm slow}/\bar{J}_{\rm fast}$ is the ratio of mean pulsar current in $[0.40, 0.85]$ s to $[0.05, 0.40]$ s. The 95% net-growth bound evaluates $\max(0, 1 - 1/R_{95})$. Across all constant-B strips and $D \le 3.5$–4.0 kpc cylinders, $R < 1$ with $> 95\%$ confidence. Because $R_{\rm true} \ge 1$ is expected when no pulsars die inside $[0.40, 0.85]$ s, the observed $R < 1$ is a diagnostic of model misspecification (e.g., flux-luminosity dimming or residual B-field mixing) rather than evidence of net annihilation; bounds on $f_{\rm slow}^{\rm exit}$ derived from R therefore carry this caveat.

Table 6. Summary of Slow-Birth Estimands, Point Estimates, 95% Upper Bounds (Rounded to 2 Sig. Figures), and Dominant Systematics

Method / Protocol	Target Estimand	Sample (N)	Point Est.	95% Upper Bound	Dominant Systematic Limitation
Primary 2D Weak-Form Continuity Inversion (CL_s Calibrated Across 300 Seeds/Dose)					
P1: Fixed Gauge	$f_{\rm slow}^{\rm birth}$	ATNF (2, 356)	0.57%	$\le 25\%$	Source-grid aliasing relative to the test window
P2: Refit Matched Gauge	$f_{\rm slow}^{\rm birth}$	ATNF (2, 356)	0.57%	$\le 28\%$	Assumes visibility lies in hinged log-quadratic family
P3: Softplus Misspec.	$f_{\rm slow}^{\rm birth}$	ATNF (2, 356)	0.57%	$\le 27\%$	Functional mismatch in 2D polar-cap dimming law
P4: Death-Line Jitter	$f_{\rm slow}^{\rm birth}$	ATNF (2, 356)	0.57%	$\le 30\%$	Scatter in death-line intercept, width, and slope
Parametric Gaussian	$f_{\rm slow}^{\rm birth}$	ATNF (2, 356)	0.00%	$\le 34\%$	Unimodal Gaussian underfits localized bimodal bumps
External 3D FGK06	$f_{\rm slow}^{\rm birth}$	Synth (2, 356)	14%–16% floor	$\le 33\%$–41% apparent	Non-separable 3D (l, b, D, DM) selection coupling
Independent Gauge-Free Observational Cross-Checks (Tests 1–4)					
Test 1: SNR Census	$f_{\rm slow}^{\rm birth}$	Radio SNRs (20)	0% (0/20)	$\le 22\%$ (raw; $\le 14\%$ back-integrated)	Small sample ($N = 20$); SNR expansion-age uncertainty
Test 2: $\lvert z\rvert$ Clock (`pksmb`)	$f_{\rm slow}^{\rm window}$	$\lvert b\rvert \le 5^\circ$ (971)	0.0% (2.6σ)	$\le 17\%$ (16.8%; $\le 3\%$ high-B*)	YMW16 distance errors; $\lvert b\rvert \le 5^\circ$ latitude footprint
Test 2: $\lvert z\rvert$ Clock ($D \in [1, 4]$ kpc)	$f_{\rm slow}^{\rm window}$	Annulus (944)	0.0% (2.6σ)	$\le 24\%$	Mixes survey latitude footprints within $[1, 4]$ kpc
Test 3: Cylinders ($D \le 4$ kpc)	Net Current Growth	Local (347–412)	$R = 0.56$–0.62	$\le 4\%$–9% ($\le 19\%$ at 3 kpc)	YMW16 distance scatter and beaming-law slope
Test 4: Strip Bound ($s \le 0.85$)	$f_{\rm slow}^{\rm exit}$	$\log_{10} B \in [11.8, 12.6]$	$R_{\rm obs} = 0.60$	$\le 23\%$ (vacuous $s \approx 3.1$)	Degenerate with flux dimming slope s ($\le 30\%$ at $+1\sigma$)

Note. All upper bounds are rounded to two significant figures (*denotes boundary-clamped active-strip bound without frequentist coverage guarantee). For Protocol P2, the canonical window yields $\le 28\%$, whereas shifted and widened window definitions yield tighter bounds of 19.4%–21.1% (Section 3.5). Each estimand ($f_{\rm slow}^{\rm birth}$, $f_{\rm slow}^{\rm window}$, $f_{\rm slow}^{\rm exit}$) is defined in Equation (3).

benchmarks (Cases A–C) test internal self-consistency rather than absolute accuracy under model error; the 3D FGK06 benchmark in Section 5.3 provides the out-of-distribution test. Second, a *domain-edge sensitivity*: the left-boundary cell at $P_0 = 35$ ms ($x_0 = -1.45$) carries 26.0% of the recovered marginal weight; as shown in Table 2, shifting the domain edge across $\log_{10} P_{\rm min} \in [-1.75, -1.35]$ keeps the fast peak confined to $P_0 \in [43.1, 59.8]$ ms and slow births $\le 1.1\%$, but redistributes weight between the boundary bin (5.9%–83.3%) and interior cells, indicating that the fast peak carries an edge-discretization uncertainty and genuinely extends below 50 ms. Third, a *goodness-of-fit check*: at the $\chi^2_{\rm min} = 192.9$ minimum ($h_0 = 0.005$ Myr^{-1}), the reduced chi-squared over the $\nu_{\rm eff} = 491.7$ residual degrees of freedom (Section 3.3) is $\chi^2/\nu_{\rm eff} = 0.392 < 1$; this sub-unity value is driven by the conservative diagonal jackknife moment variances and empirical variance floor in the weak-form preconditioning, coupled with the positive spatial correlation among overlapping test functions, rather than unmodeled over-regularization. Forward Kolmogorov–Smirnov distances $D_{\rm KS}(P) = 0.043$ and $D_{\rm KS}(B) = 0.036$ ($p \approx 0.05$–0.20) confirm agreement with the observed 1D marginal distributions.

Applied to $N = 2,356$ canonical isolated pulsars in the ATNF Pulsar Catalogue (Table 6 and Figure 7b), the 2D inversion yields $\hat{f}_{\rm slow}^{\rm birth} = 0.57\%$ ($\le 9.03\%$ across $h_0 \in [0, 0.048]$ Myr^{-1}, with 94.5% of births at $P_0 \le 0.21$ s and Galactic birthrate $\mathcal{R}_{\rm birth} = 1.82 \pm 0.45$ century^{-1} at baseline (the 300-seed Monte Carlo ensembles at 0% injection span 1.95–2.18 century^{-1} across Protocols P1–P4, reflecting gauge-refitting and death-line variance), concordant with the Milky Way core-collapse supernova rate). This result is independently corroborated by four gauge-free cross-checks: (1) a supernova remnant birth-period census (1/20 radio SNR pulsars in $[0.40, 0.85]$ s without spin-down correction, $f_{\rm slow}^{\rm birth} \le 22\%$; tightening to $\le 14\%$ under canonical $t_{\rm SNR} = 35$ kyr back-integration for B2334+61), (2) an unweighted vertical Galactic scale-height clock $\lvert z\rvert = D\lvert\sin b\rvert$ (2.6σ tension with a 50% slow-birth mixture and $f_{\rm slow}^{\rm window} \le 17\%$ in the uniform Parkes Multibeam $\lvert b\rvert \le 5^\circ$ footprint; $\le 24\%$ in $D \in [1, 4]$ kpc), (3) doubly truncated volume-complete cylinders ($D \le 3.5$–4.0 kpc, $R = 0.56$–$0.62 < 1$, bounding net current growth to $\le 4\%$–9%) and dis-

joint constant-B strips ($R = 0.28$–$0.66 < 1$), and (4) a high-B strip conservation inequality ($f_{\rm slow}^{\rm exit} \leq 23\%$ for dimming slopes $s \leq 0.85$, and $\leq 30\%$ at $+1\sigma$).

Taken together, our 300-seed CL_s calibration and external 3D benchmarking establish a clear boundary between what current pulsar catalogues exclude and what remains unresolved. The classical hypothesis that $\geq$ 40% of pulsars are injected at $P_0 \in [0.40, 0.85]$ s (Vivekanand & Narayan 1981) is excluded at $> 95\%$ confidence across every internal protocol ($\mathrm{CL}_s = 0.000$ at 40% dose), every gauge-free cross-check, and the external 3D FGK06 benchmark (98% rejection power at 40% dose). Conversely, because regularized non-negative inversion over a 2D flux-limited catalogue has finite resolution ($\mathrm{FWHM}_x \approx 0.21$–$0.62$ dex) and non-separable 3D spatial-kinematic-radiometer selection coupling introduces a $\sim$ 14%–16% baseline floor on external 3D synthetic catalogues, slow-birth fractions below 25%–30% under our 2D protocols—and apparent fractions below 33%–41% under 3D selection mismatch—cannot be excluded with current data. Breaking this remaining resolution floor will require homogeneous wide-band flux measurements and VLBI/SKA pulsar parallaxes across the Galactic plane to replace 2D (P, B) visibility gauges with direct 3D (P, B, D, DM) forward operators.

7.1. *Scientific Impact and Methodological Significance*

This work makes both a methodological contribution to inverse problems in physics and a quantitative contribution to the study of pulsar birth demographics. Its principal methodological innovation is a weak-form, two-dimensional continuity inversion in $(\log_{10} P, \log_{10} B)$ that reconstructs the underlying birth distribution without numerically differentiating sparse catalogue data. By transferring phase-space derivatives onto smooth test functions, the method reduces sensitivity to sampling noise while retaining the coupled effects of magnetic field and rotational evolution. Calibrating the visibility function outside the target birth-period interval limits circularity in the inference, while the resolution matrix characterises the inversion's resolving power. Monte Carlo dose-response experiments and calibrated frequentist exclusion limits further quantify the extent to which a putative slow-birth population can be distinguished from the zero-injection baseline. Together, these features provide a systematic way to distinguish a small reconstructed population from one that can be statistically excluded.

Applied to 2,356 isolated pulsars in the ATNF Catalogue, the inversion yields a slow-birth fraction of 0.57% in the 0.40–0.85 s initial-period interval, with point estimates remaining below 9.03% across the examined magnetic-field decay rates. The reconstructed birth-period distribution is concentrated at initial periods below 0.21 s. More importantly, the calibrated 95% upper limits of approximately 25–30% across the primary protocols, together with four independent observational cross-checks, disfavor the classical hypothesis that at least 40% of pulsars are born in the specified slow-period interval. The analysis also demonstrates that a pronounced feature in the one-dimensional pulsar current need not correspond to a substantial slow-born population: the two-dimensional spin-down dynamics can reproduce such a feature even when the reconstructed births are concentrated at short periods. This provides a different interpretation of a longstanding demographic argument without requiring the complete absence of slow-born pulsars.

The broader significance of the method lies in its treatment of a general difficulty in physics: inferring initial conditions from an observed population whose members have evolved under known dynamics and whose detectability is imperfect. The combination of conservation equations, regularised inversion, explicit resolution analysis, and calibrated hypothesis testing offers a framework that may be adapted to other physical systems when their evolution and observation processes can be modelled appropriately. However, the results also establish important limits on the present inference. Slow-birth fractions below approximately 25–30% cannot be excluded under the primary protocols, and the independent three-dimensional benchmark reveals additional biases when spatial and survey-selection effects are not fully represented by the two-dimensional visibility model. Improved homogeneous wide-band flux measurements and pulsar distance constraints, including very-long-baseline interferometry parallaxes and future SKA observations, will therefore be important for constructing more realistic three-dimensional forward models and testing the reconstructed birth distribution with greater statistical power.

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